23CSE465 MOBILE APPLICATION DEVELOPMENT

CHARGEIQ – EV CHARGING LOCATOR

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INTRODUCTION

The rapid growth of electric vehicles (EVs) has increased the demand for efficient and accessible charging infrastructure. However, EV users often face difficulties in locating available charging stations, estimating waiting times, and selecting the most suitable station based on cost, distance, and charging speed. These challenges create inconvenience and contribute to range anxiety among EV drivers.

ChargeIQ is a smart EV charging management application designed to simplify the process of finding and using charging stations. The system enables users to locate nearby EV charging stations, monitor real-time availability, and make optimized charging decisions using data-driven insights. By integrating location services, real-time data monitoring, and intelligent recommendation mechanisms, ChargeIQ aims to enhance the charging experience for EV users and support the adoption of sustainable transportation solutions.

PROBLEM STATEMENT

As the number of electric vehicles continues to grow, the availability and accessibility of charging infrastructure become critical factors for EV adoption. Many EV users struggle to find nearby charging stations that are compatible with their vehicles and available at the time of arrival. Existing charging locator applications often provide only basic location information without considering factors such as real-time availability, charging costs, or station efficiency.

This lack of intelligent charging guidance can lead to inefficient route planning, longer waiting times, and underutilization of charging infrastructure. Drivers may arrive at charging stations that are already occupied or unsuitable for their charging needs. Therefore, there is a need for an intelligent system that not only locates EV charging stations but also analyzes multiple parameters such as distance, availability, and cost to recommend the most optimal charging option for users.

ChargeIQ addresses this problem by providing a smart platform that helps EV drivers locate charging stations and make informed decisions using real-time data and intelligent recommendations.

OBJECTIVES OF THE PROJECT

The primary objective of the ChargeIQ system is to develop a smart and efficient platform that assists EV users in locating and selecting optimal charging stations. The specific objectives of the project include:

- To develop an application that locates nearby EV charging stations using location-based services.
- To provide real-time information about charging station availability and status.
- To recommend the most suitable charging stations based on factors such as distance, cost, and availability.
- To improve the overall charging experience for EV users through intelligent decision support.
- To support the growing EV ecosystem by making charging infrastructure easier to access and manage.

SCOPE OF THE PROJECT

The scope of the ChargeIQ project focuses on providing a smart EV charging locator and recommendation system that enhances the usability and efficiency of charging infrastructure.

The system allows users to search for nearby charging stations, view detailed station information, and receive recommendations based on multiple parameters such as location, pricing, and availability. The application also provides real-time updates about charging station status to help users make informed decisions before reaching a station.

ChargeIQ is designed to function as a scalable platform that can integrate with charging station databases and location services. While the current system focuses primarily on station discovery and intelligent recommendations, it can be extended in the future to include features such as charging slot reservations, route-based charging suggestions, user analytics dashboards, and integration with smart energy management systems.

SYSTEM ARCHITECTURE

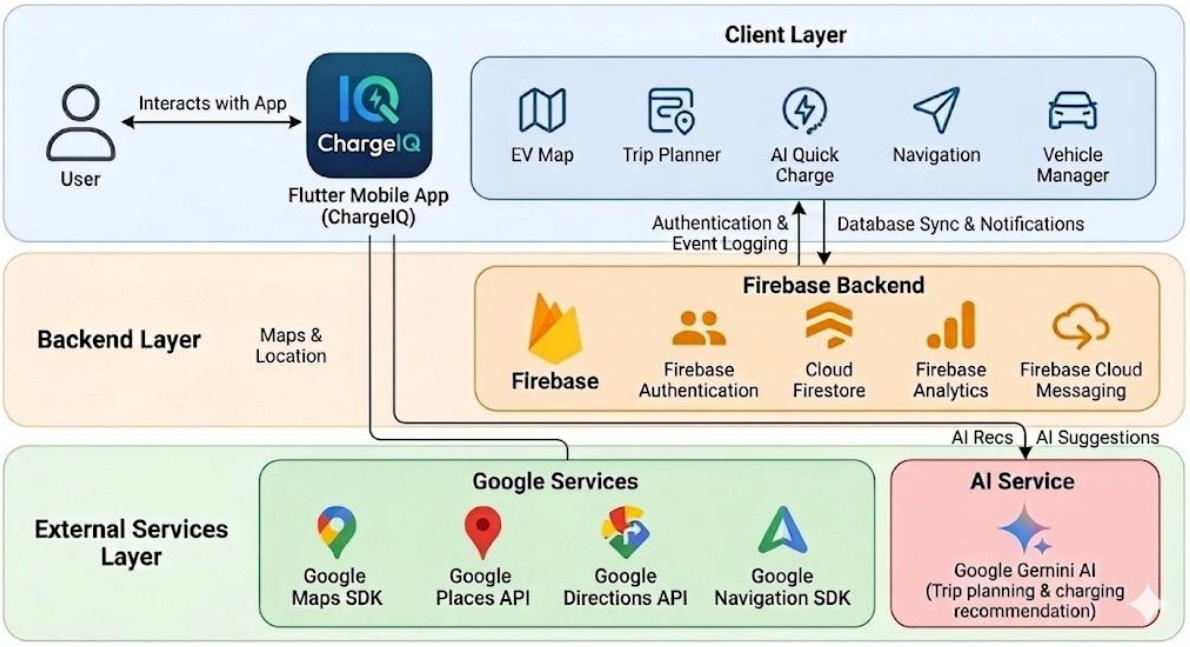

Figure 1. System Architecture of ChargeIQ

FEATURES OF CHARGEIQ

- **EV Charging Station Locator** – Displays nearby EV charging stations on an interactive map based on the user's real-time location.
- **Smart Location Search** – Allows users to search for charging stations by city or location using an autocomplete search system.
- **Real-Time Station Availability** – Shows whether a charging station is currently open along with distance and ratings.
- **AI-Based Station Recommendation** – Uses an AI scoring algorithm to recommend the most optimal charging station considering factors like distance, rating, open status, and charging ports.
- **Quick Charge Mode** – Enables fast selection of the best charging station with a single action using AI optimization.
- **Interactive Map Visualization** – Displays charging stations as map markers, enabling easy identification and navigation.
- **Charging Station Details View** – Provides detailed information about each station including address, connector type (e.g., CCS2), rating, and distance.
- **Navigation Integration** – Generates routes and turn-by-turn navigation to reach selected charging stations.
- **Nearby Station Listing** – Shows a sorted list of charging stations based on distance, rating, or availability.
- **Station Filtering Options** – Allows filtering stations by:

    - Open now
    - Top rated stations
    - All available stations

- **Vehicle Profile Management** – Users can add and manage EV details including battery capacity, range, and charging port type.
- **Trip Planner** – Plans EV journeys with optimal charging stops between the source and destination.
- **Smart Route Optimization** – Calculates routes with intermediate charging stations based on the vehicle's battery range.
- **Restaurant and Stop Recommendations** – Option to include restaurants or cafes during charging stops for convenience.
- **Charging Stop Prediction** – Identifies where the vehicle needs to recharge along the route.
- **Trip History Management** – Stores previously planned routes for quick access.
- **Saved Charging Locations** – Allows users to bookmark frequently used charging stations.
- **Personalized Charging Settings** – Includes options like driving style, AC usage, battery health, and preferred charging type.
- **Dark Mode / Light Mode Interface** – Provides theme switching for improved user experience.
- **User Dashboard** – Displays usage statistics such as routes taken, stations found, and time saved.
- **Account Management** – Allows linking Google accounts, resetting passwords, and managing user profiles.

GITHUB REPOSITORY LINK

https://github.com/lithigesh/ChargeIQ

APP RELEASES PAGE

https://github.com/lithigesh/ChargeIQ/releases/tag/v1.0.0

WHY WE COULD NOT DEPLOY THIS IN VERCEL?

The application could not be deployed on Vercel because the Google Cloud free-tier credits allowed access only to the **Maps and Navigation SDK for Android**, while web deployment

requires the **Google Maps JavaScript API with billing enabled**, which exceeded the available free credits.

USER INTERFACE SCREENSHOTS

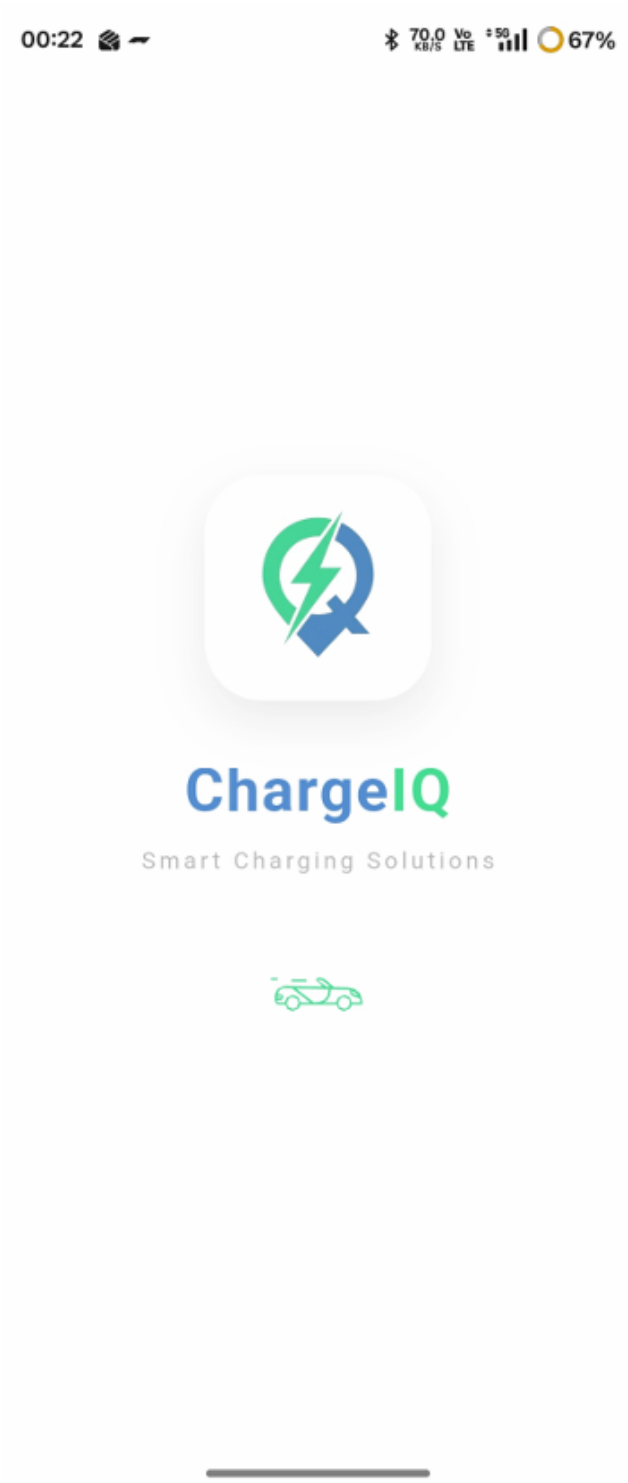

Figure 2. ChargeIQ application splash screen

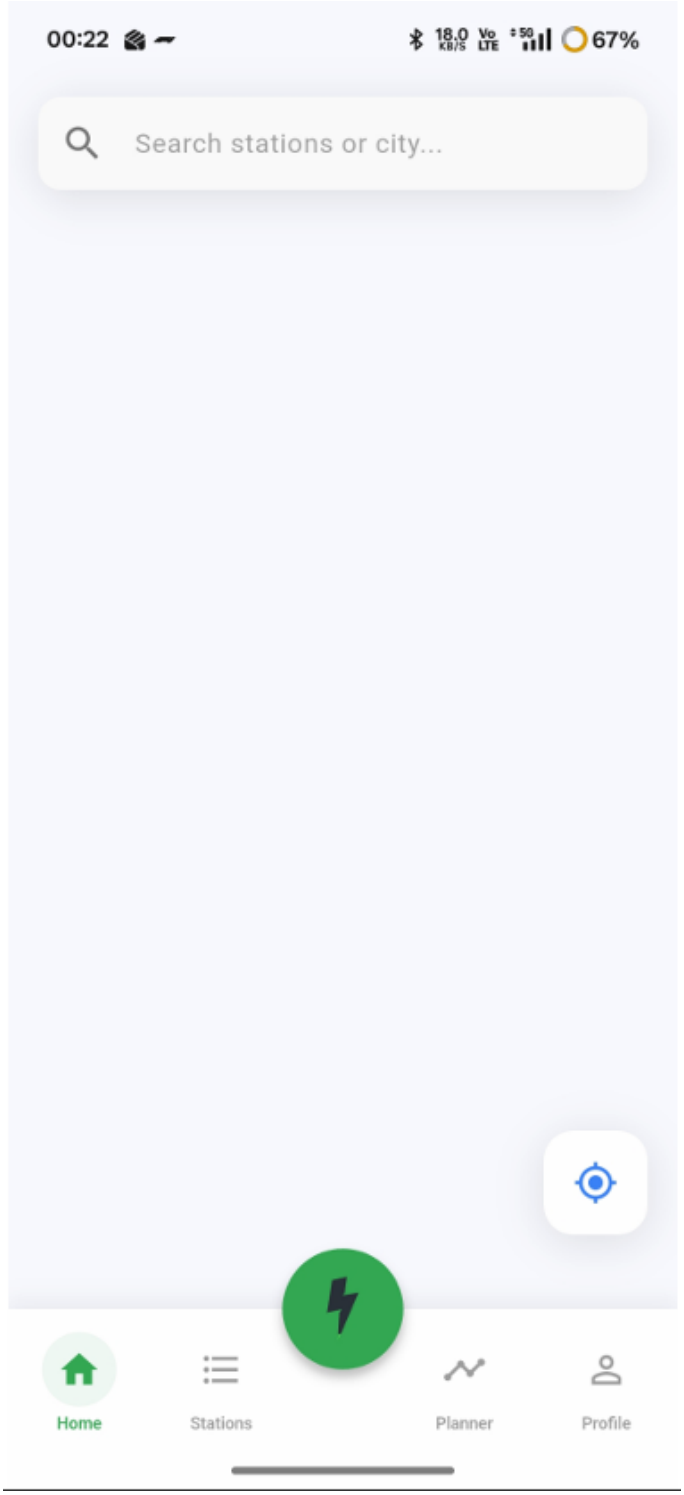

Figure 3. Home screen showing the EV charging station search interface with navigation tabs.

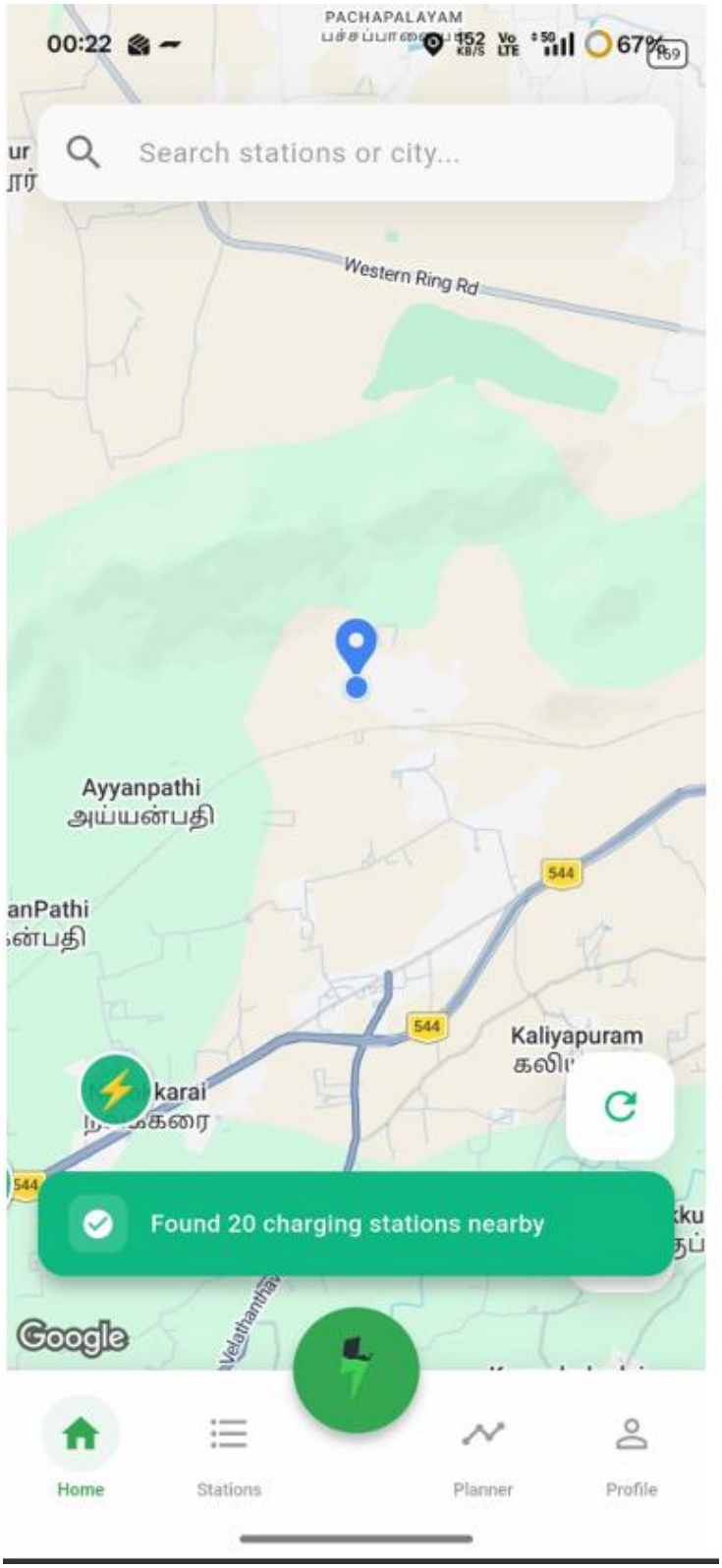

Figure 4. Home screen showing the EV charging station search interface with navigation tabs.

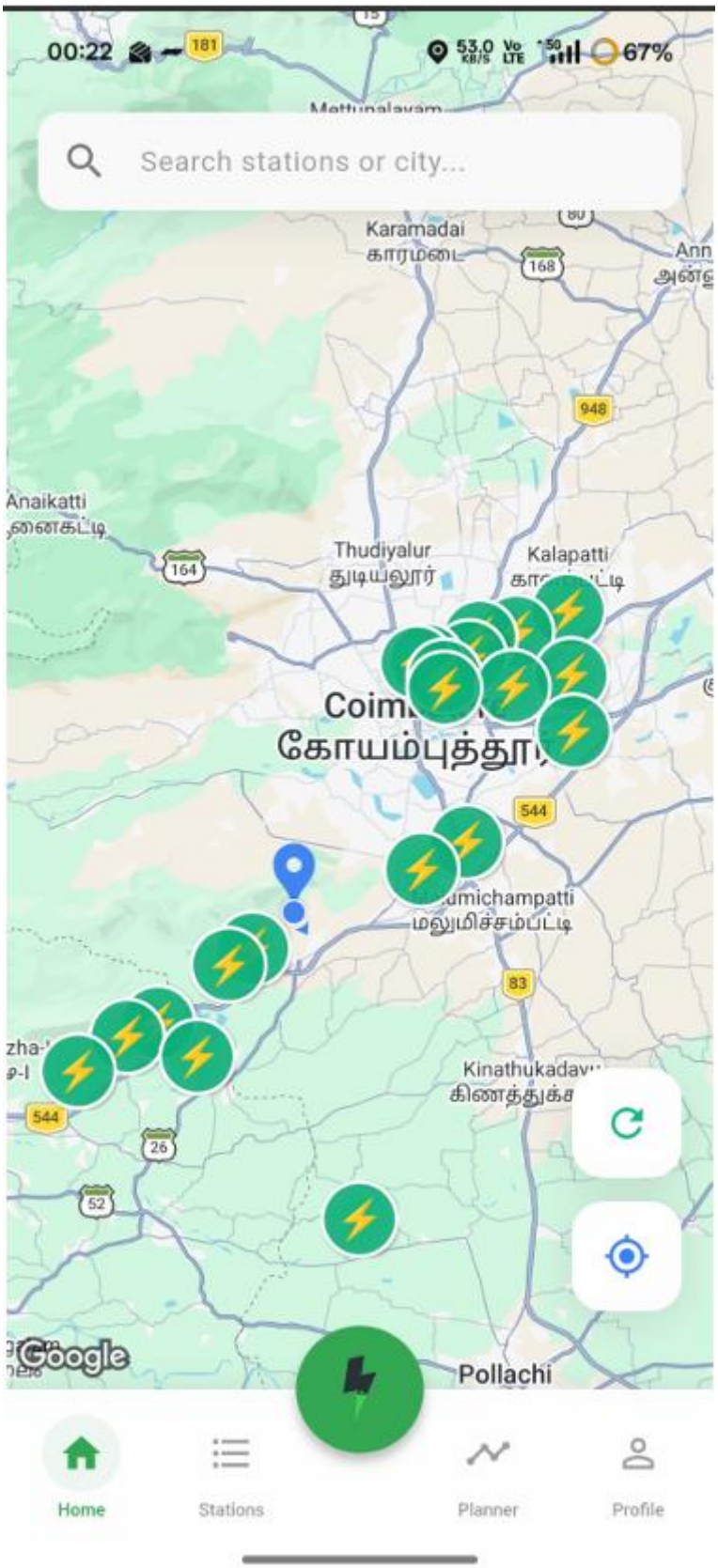

Figure 5. Multiple EV charging stations visualized on the map around Coimbatore.

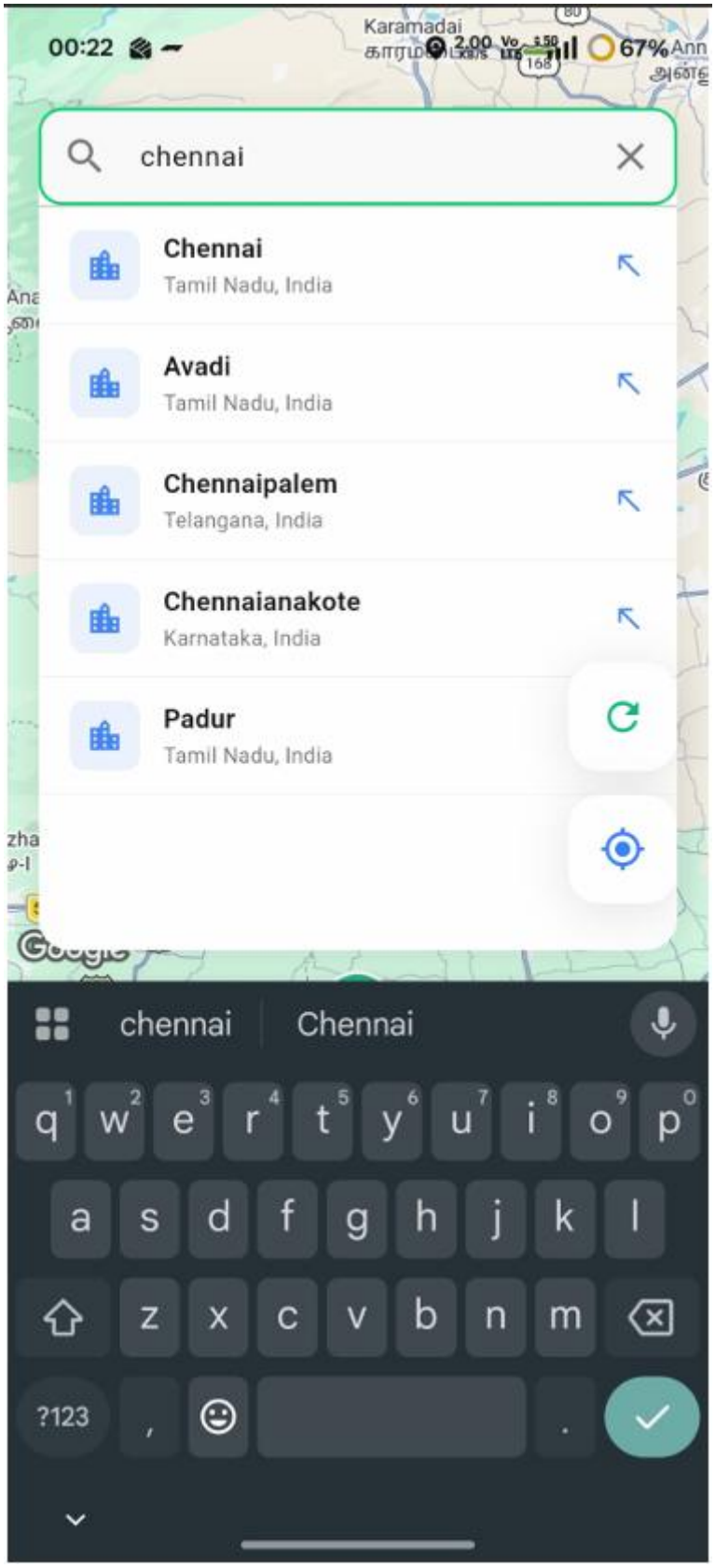

Figure 6. Search interface showing location suggestions while entering a city name.

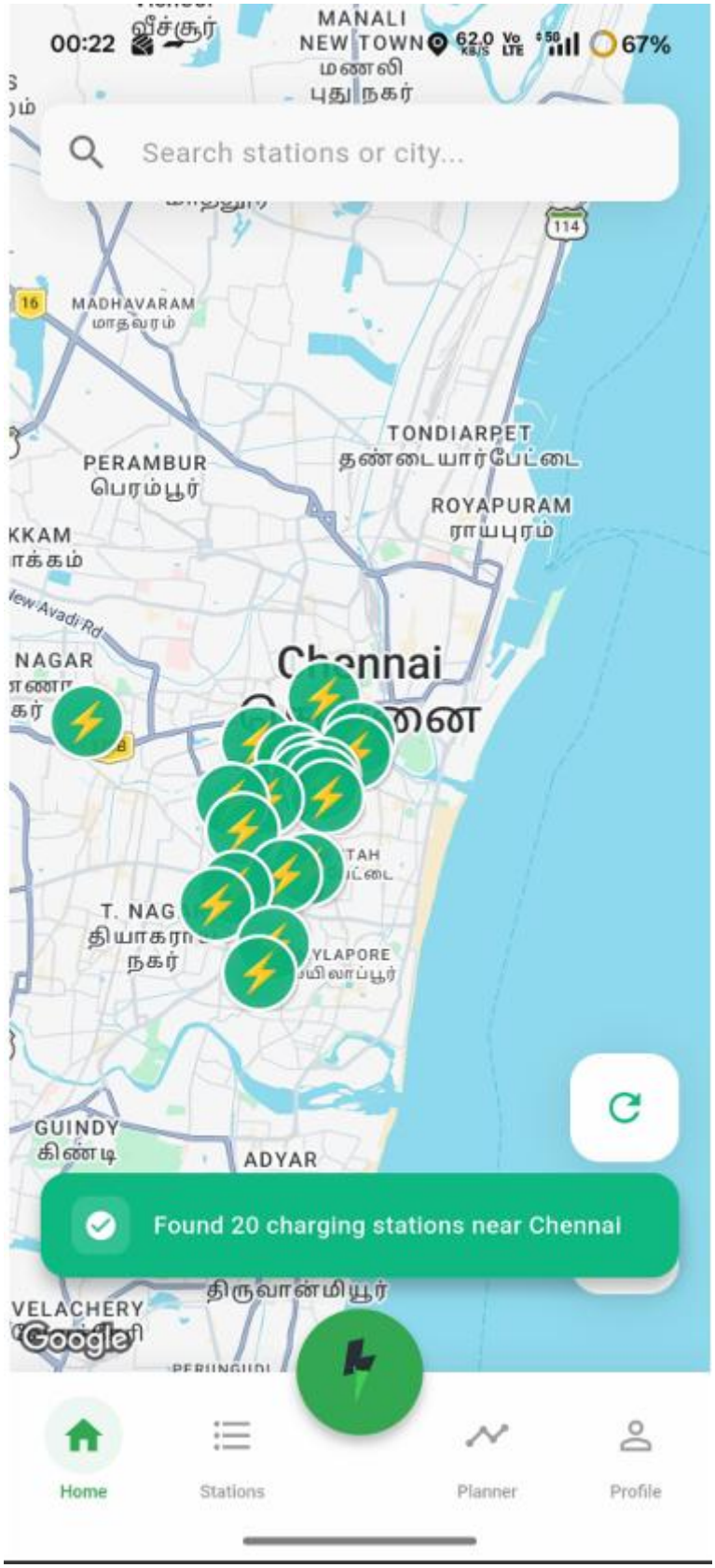

Figure 7. Map displaying EV charging stations available near Chennai.

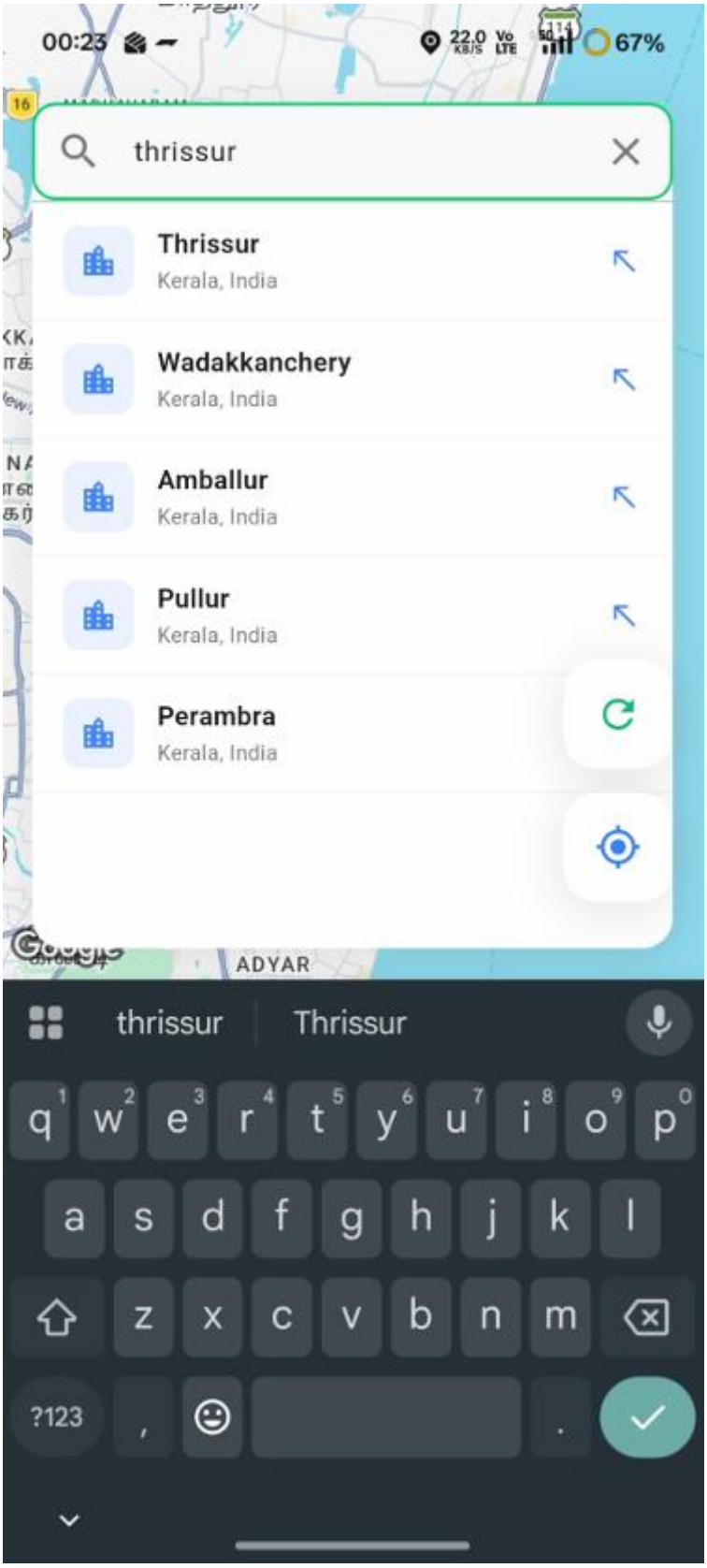

Figure 8. Location search results for Thrissur with autocomplete suggestions.

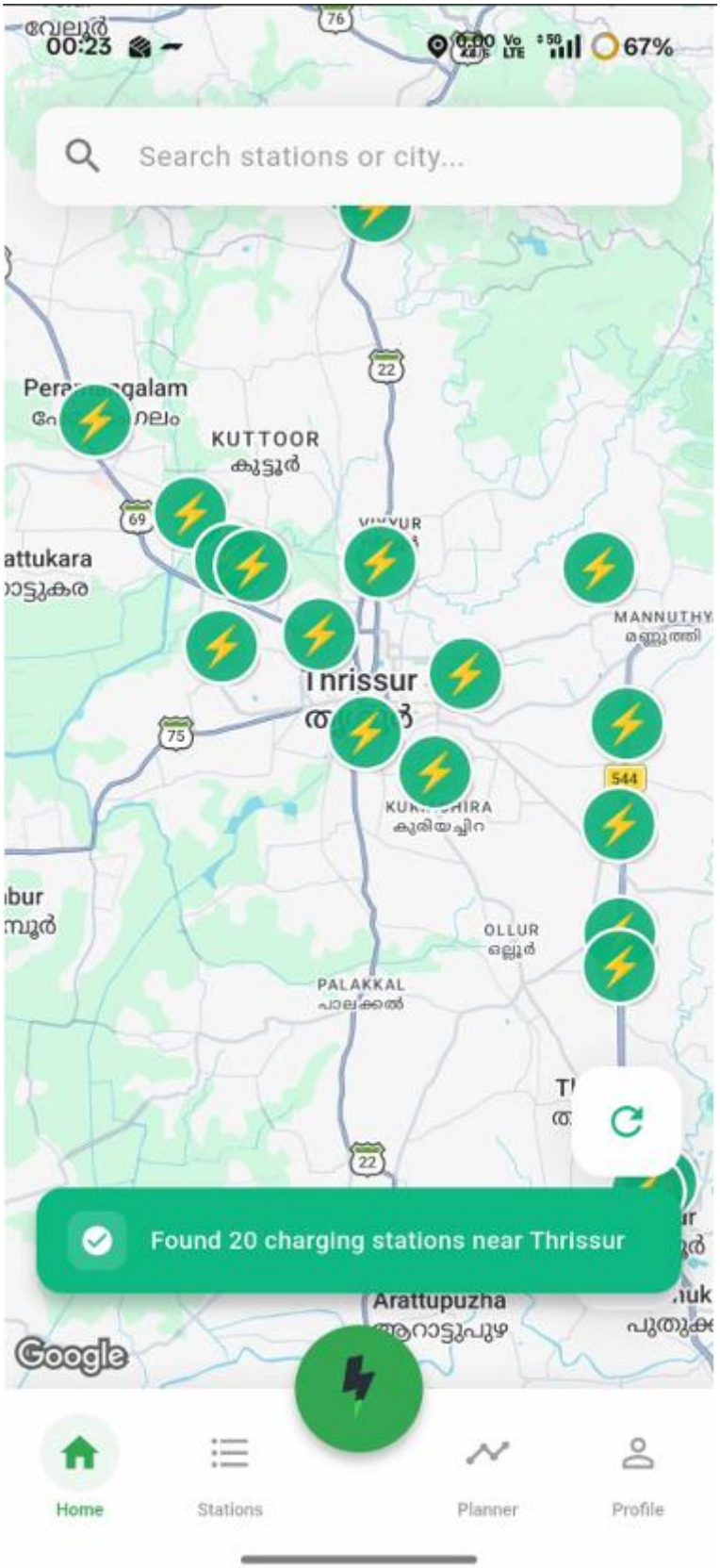

Figure 9. Map displaying EV charging stations detected near Thrissur.

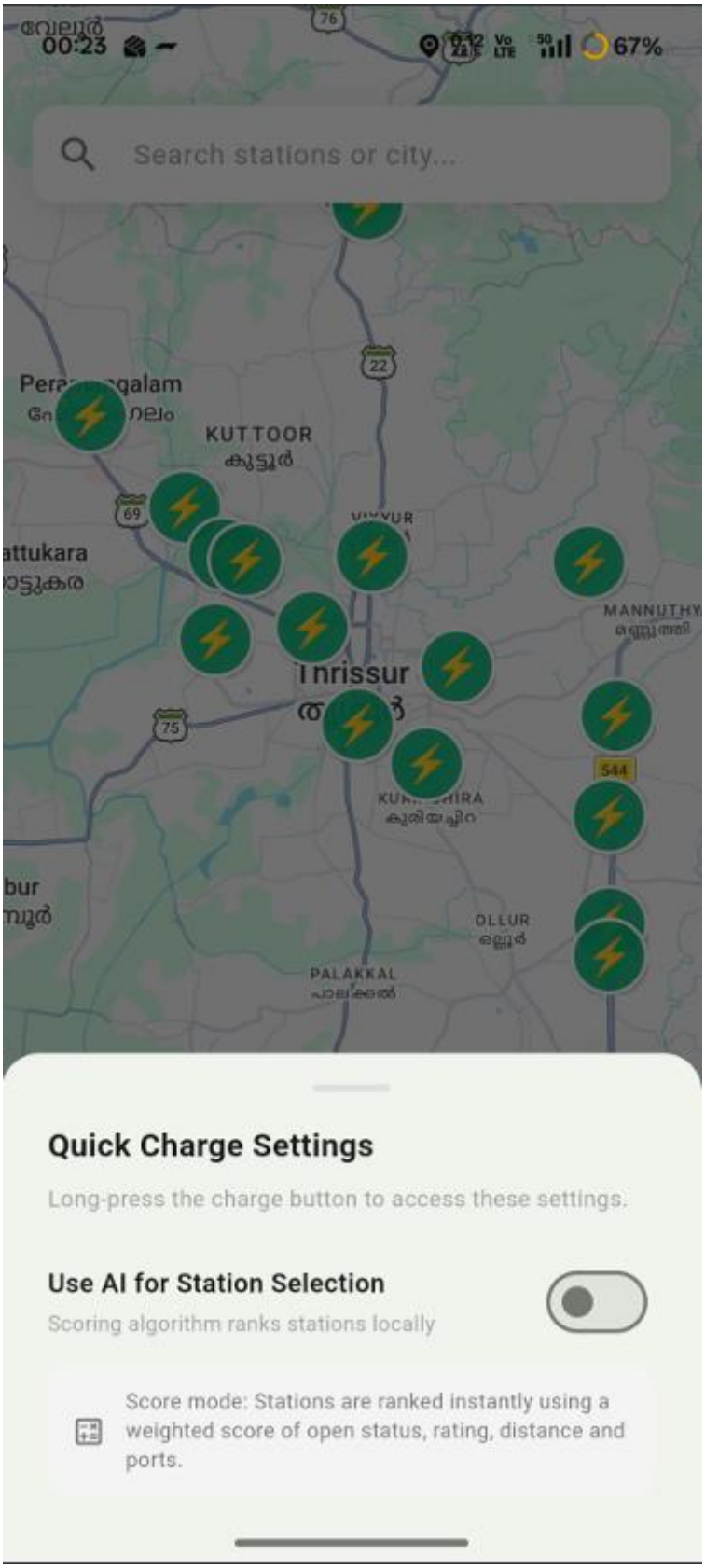

Figure 10. Quick Charge Settings panel showing the AI-based station selection option.

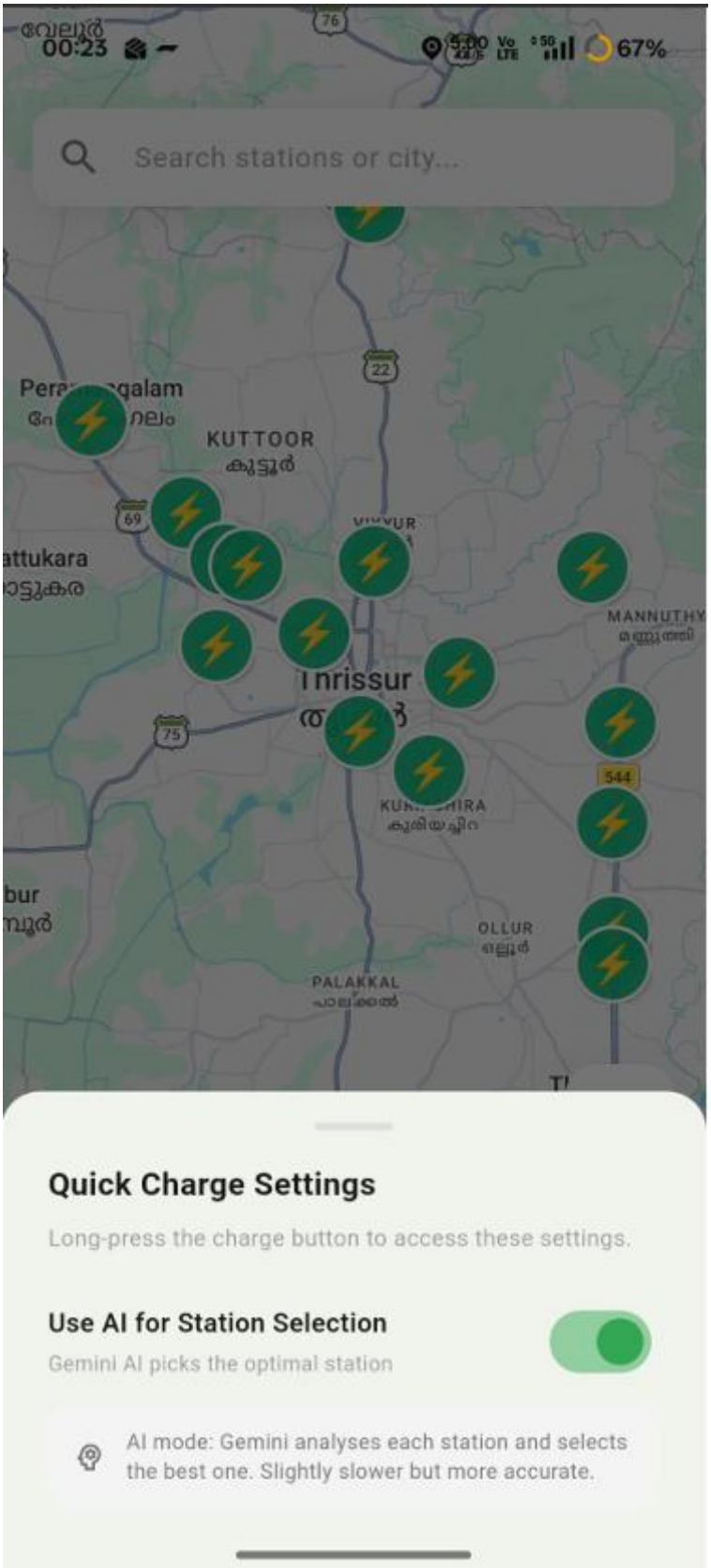

Figure 11. AI-based station recommendation enabled for optimized charging station selection.

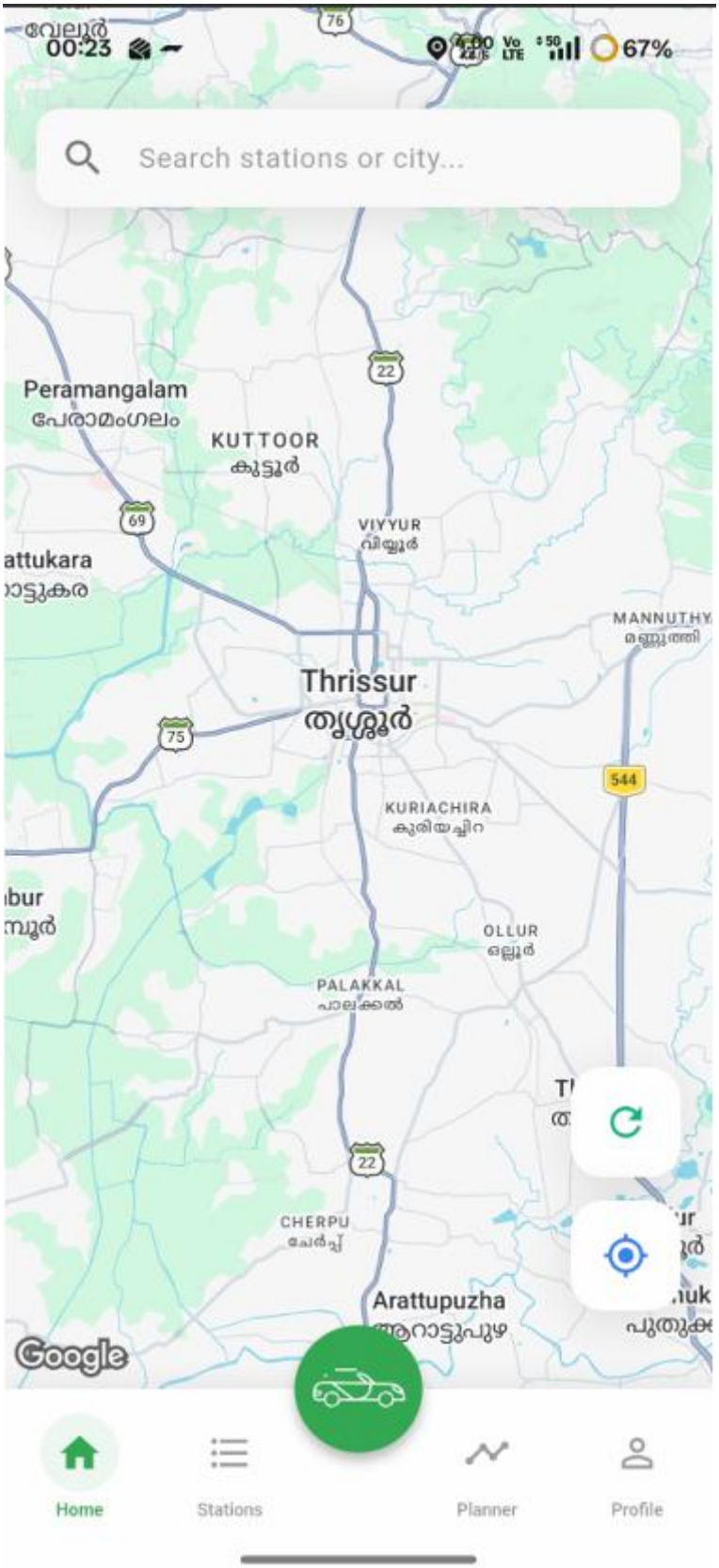

Figure 12. Map view after activating the quick charge button in AI mode

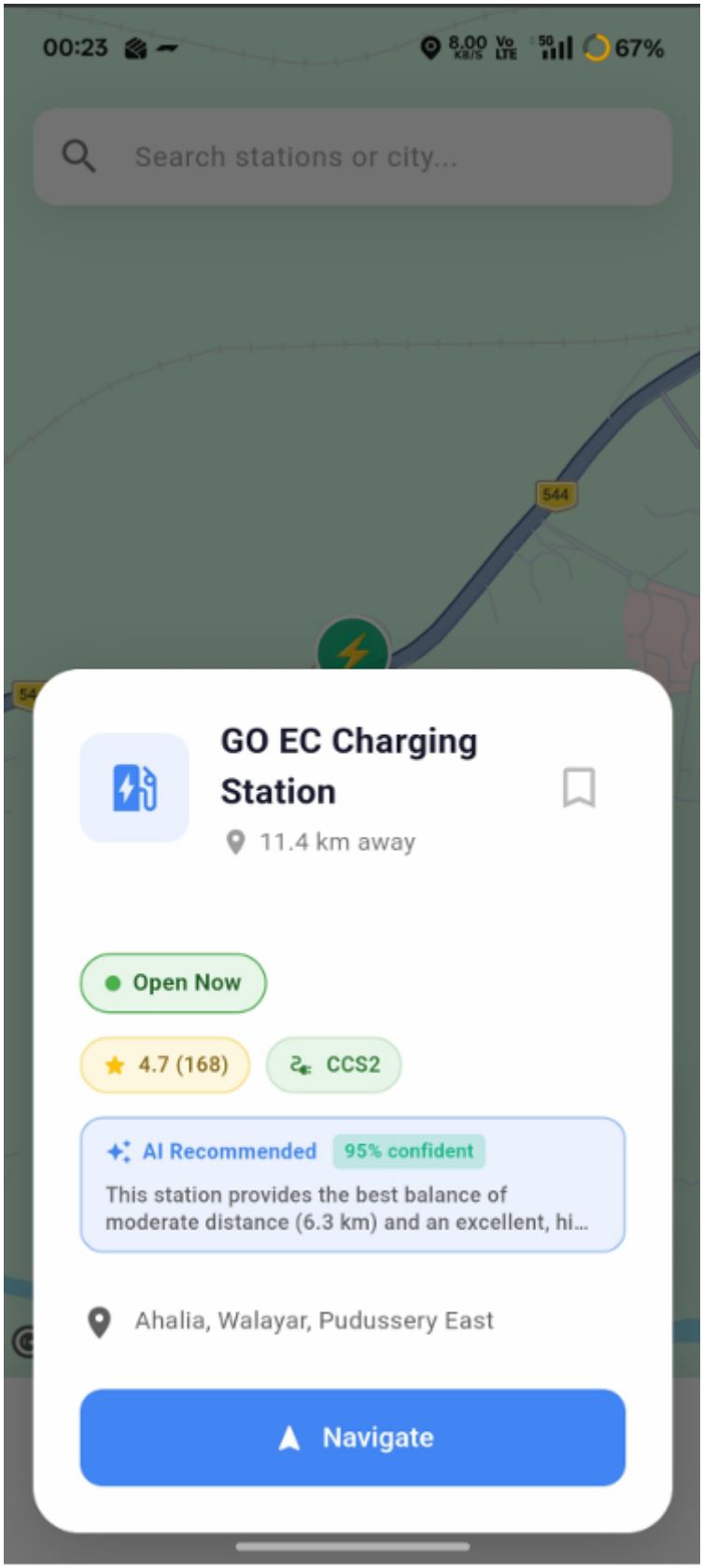

Figure 13. AI recommended quick charge station

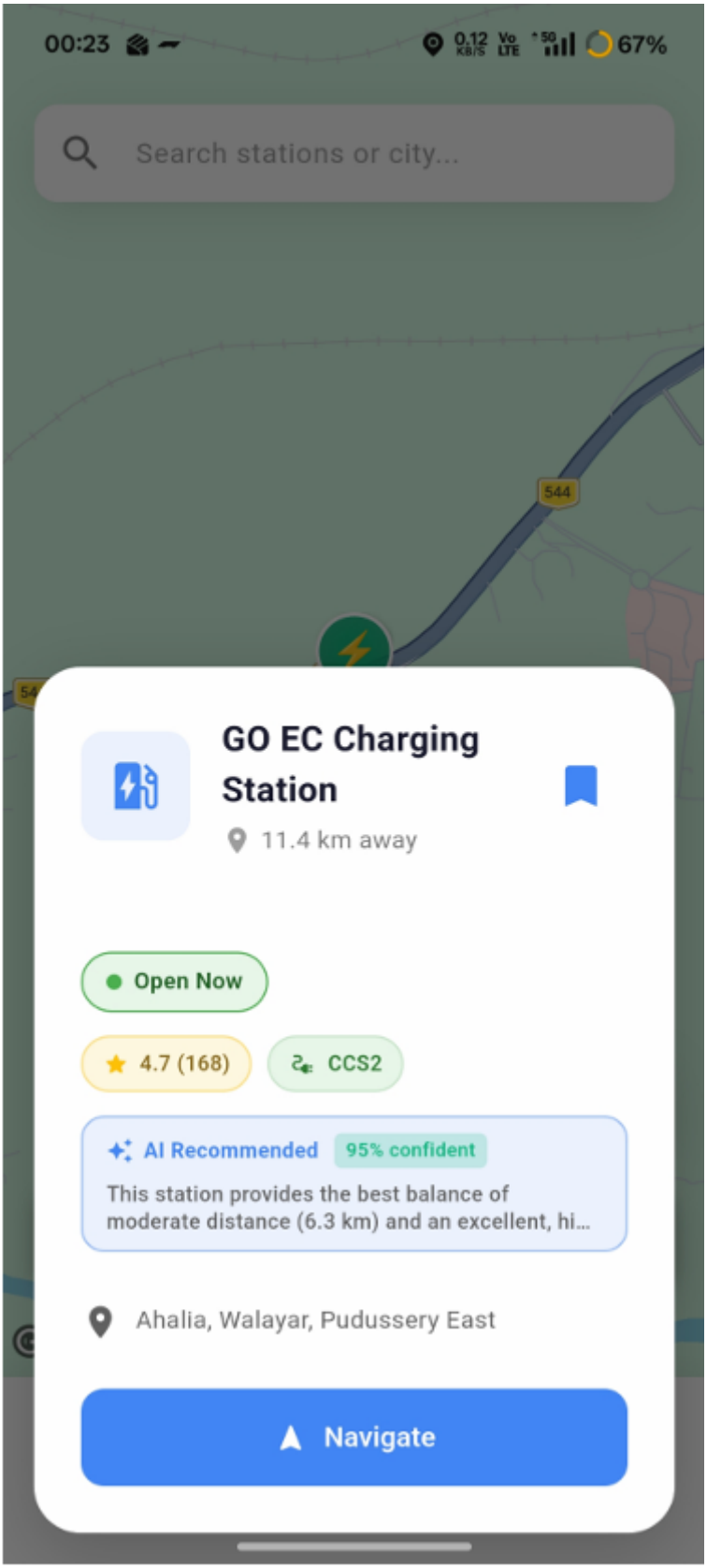

Figure 14. Charging station details with bookmark option for saving the location.

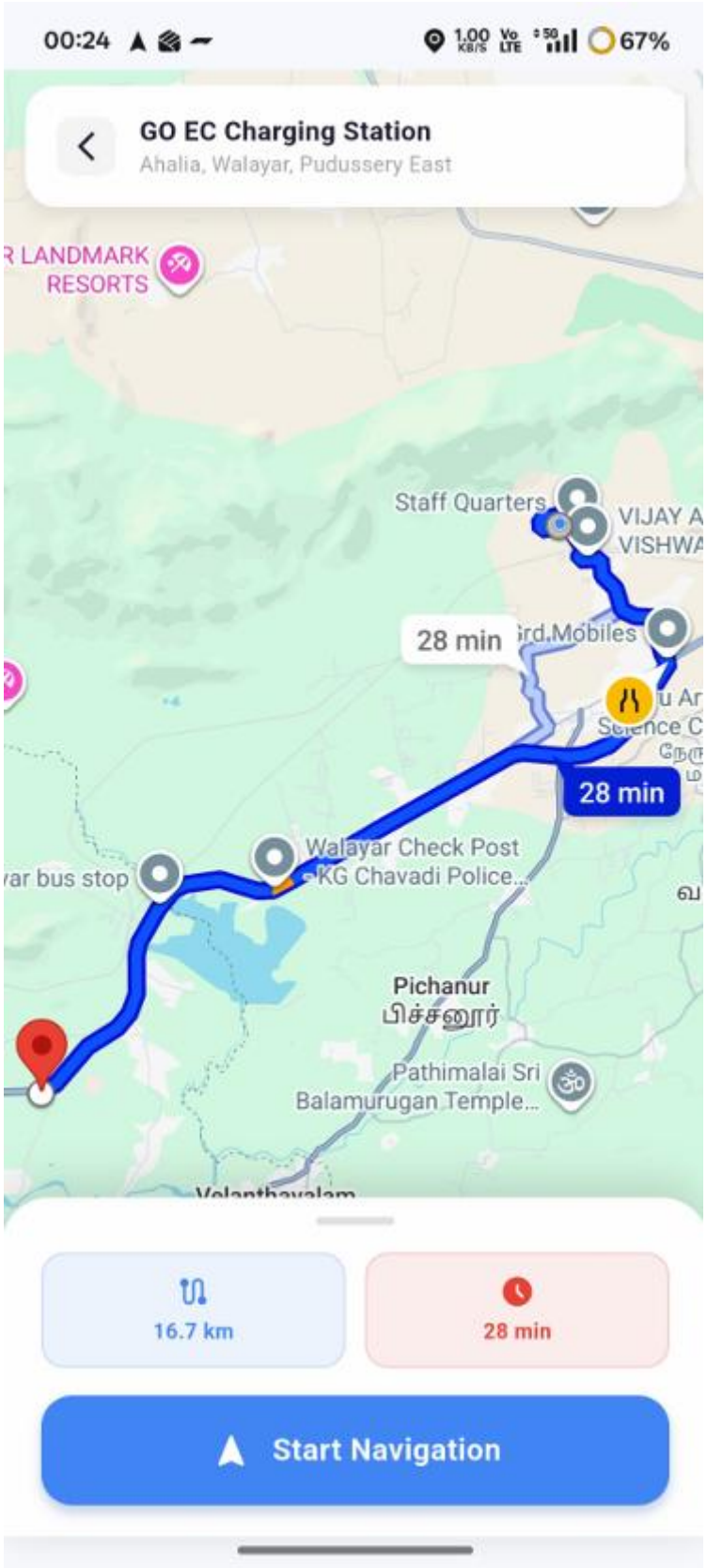

Figure 15. Route navigation interface showing the path to the selected charging station.

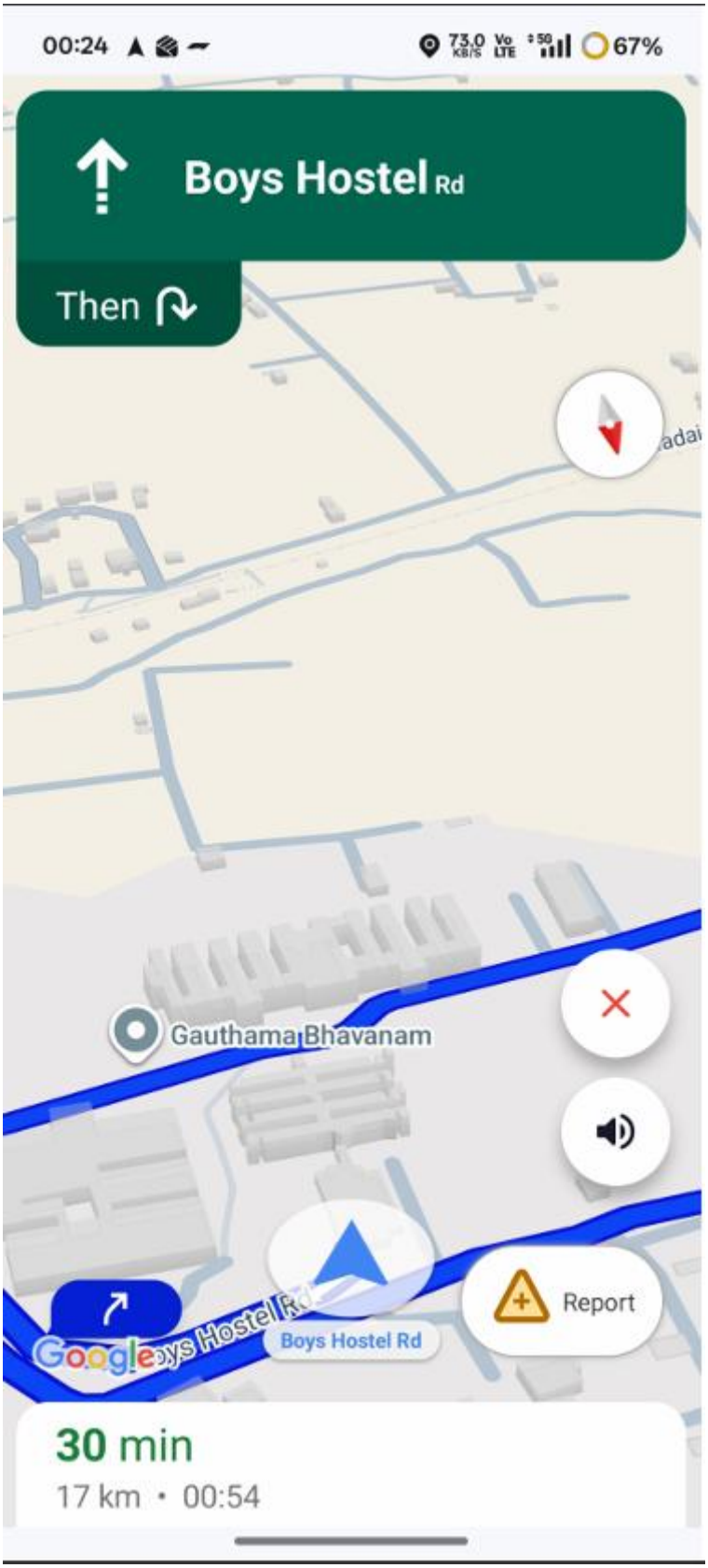

Figure 16. Turn-by-turn navigation interface displaying road instructions.

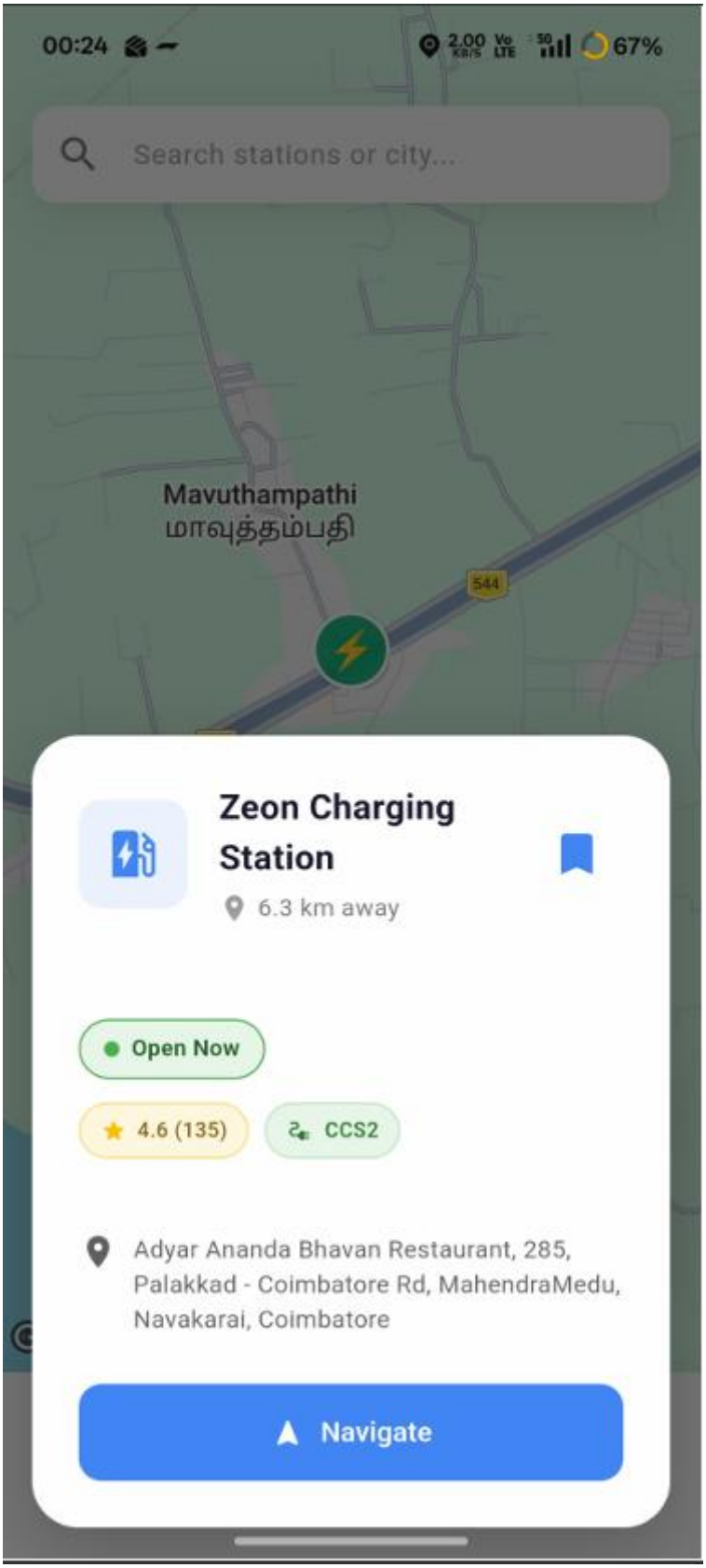

Figure 17. Scoring algorithm recommended quick charge station

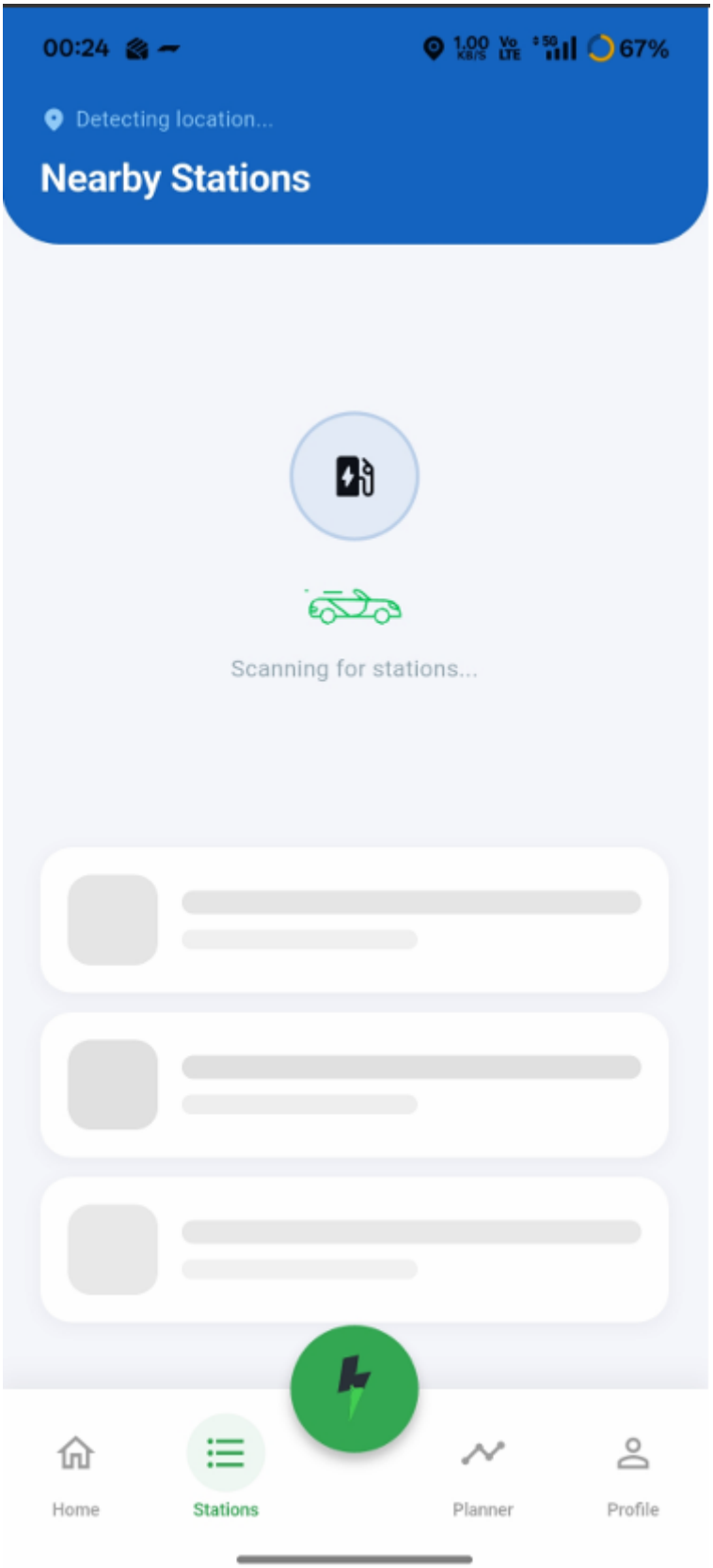

Figure 18. Nearby stations screen while scanning for available EV charging points.

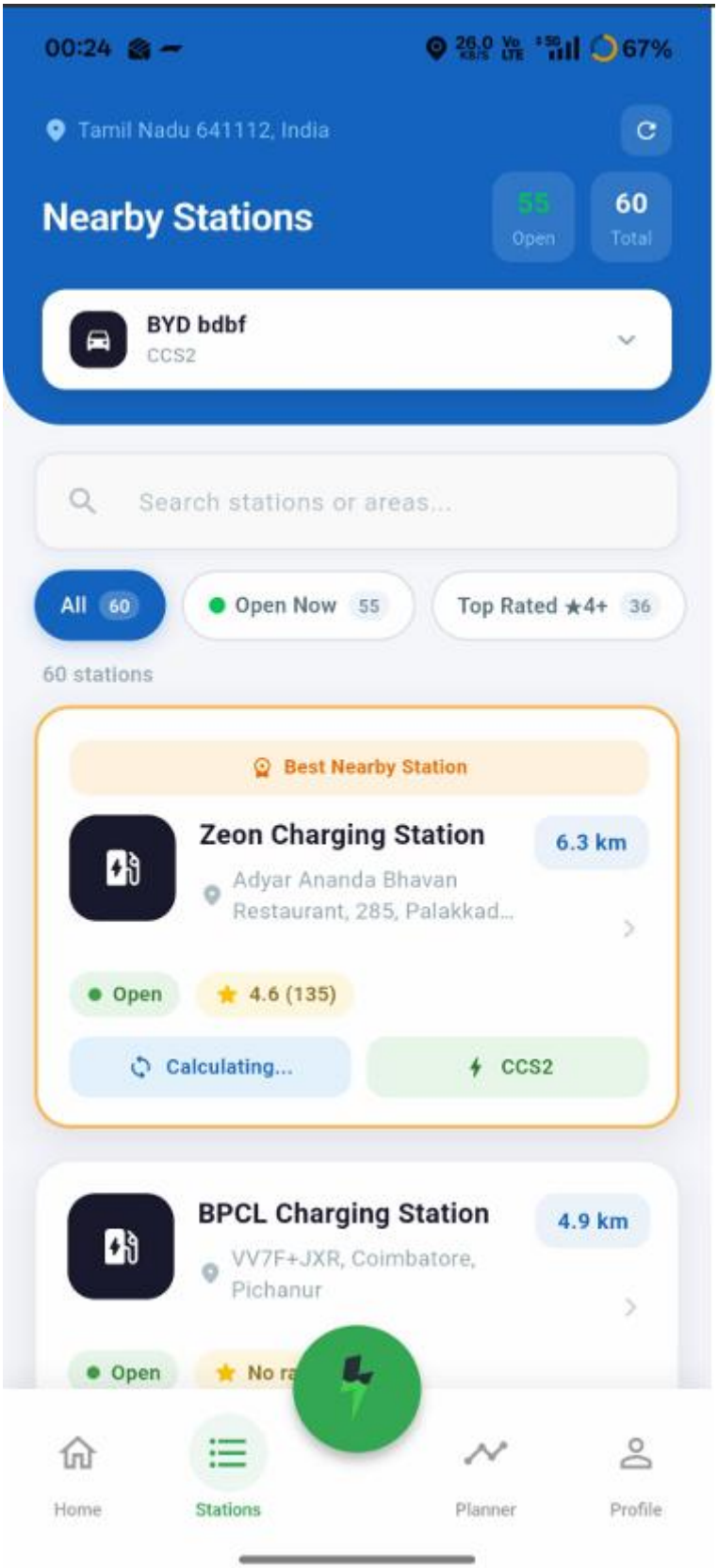

Figure 19. List of nearby charging stations with ratings and connector types.

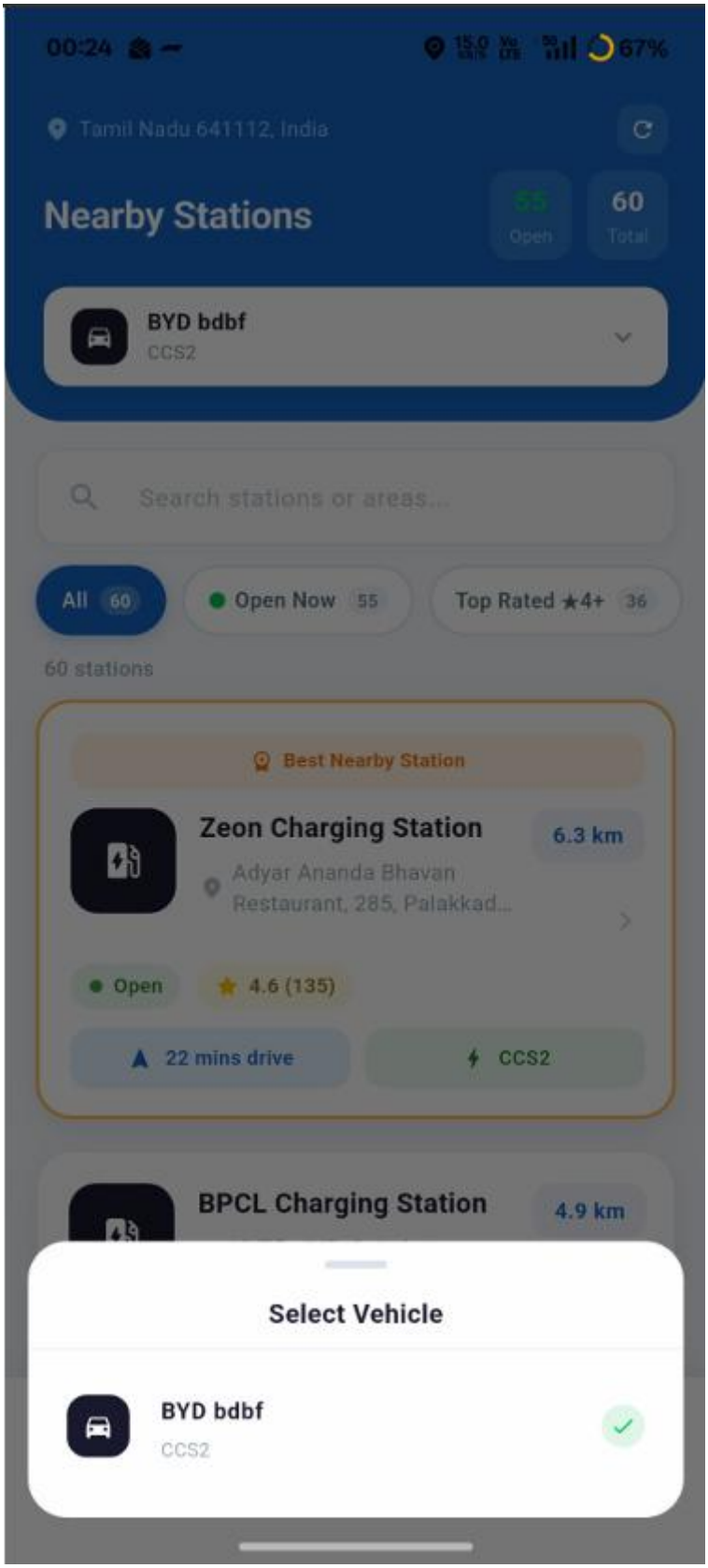

Figure 20. Vehicle selection popup for choosing the EV model.

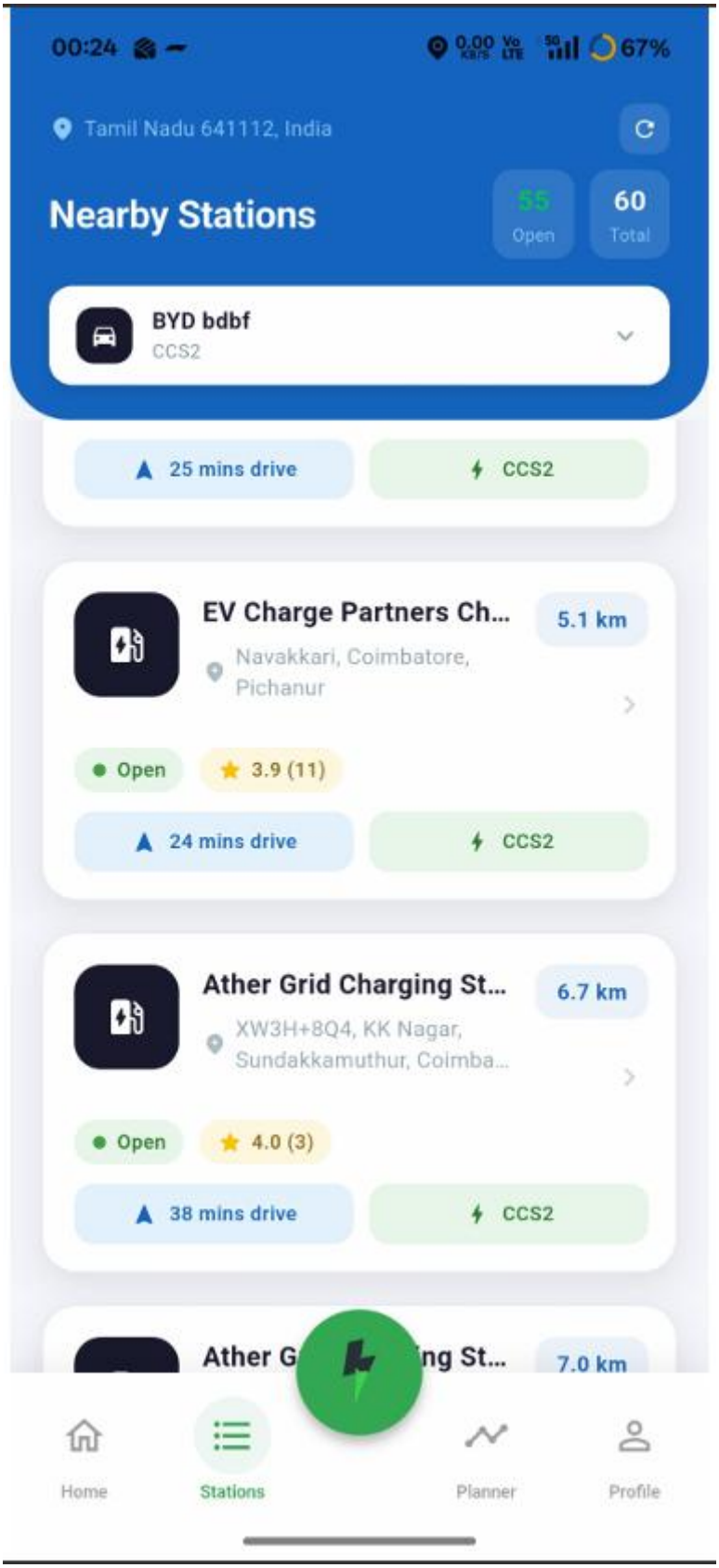

Figure 21. Detailed station list showing distance, availability, and connector type.

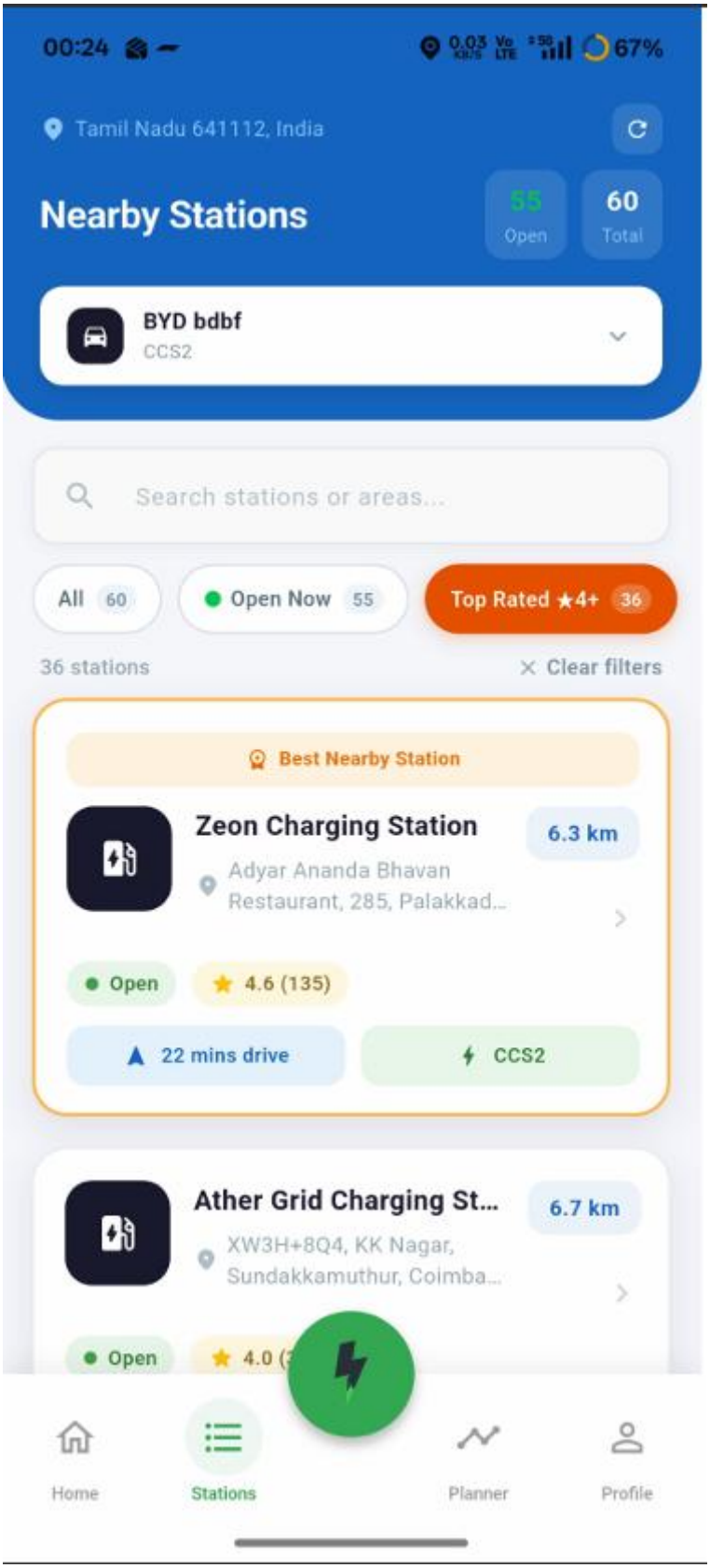

Figure 22. Filtered station list showing top-rated charging stations.

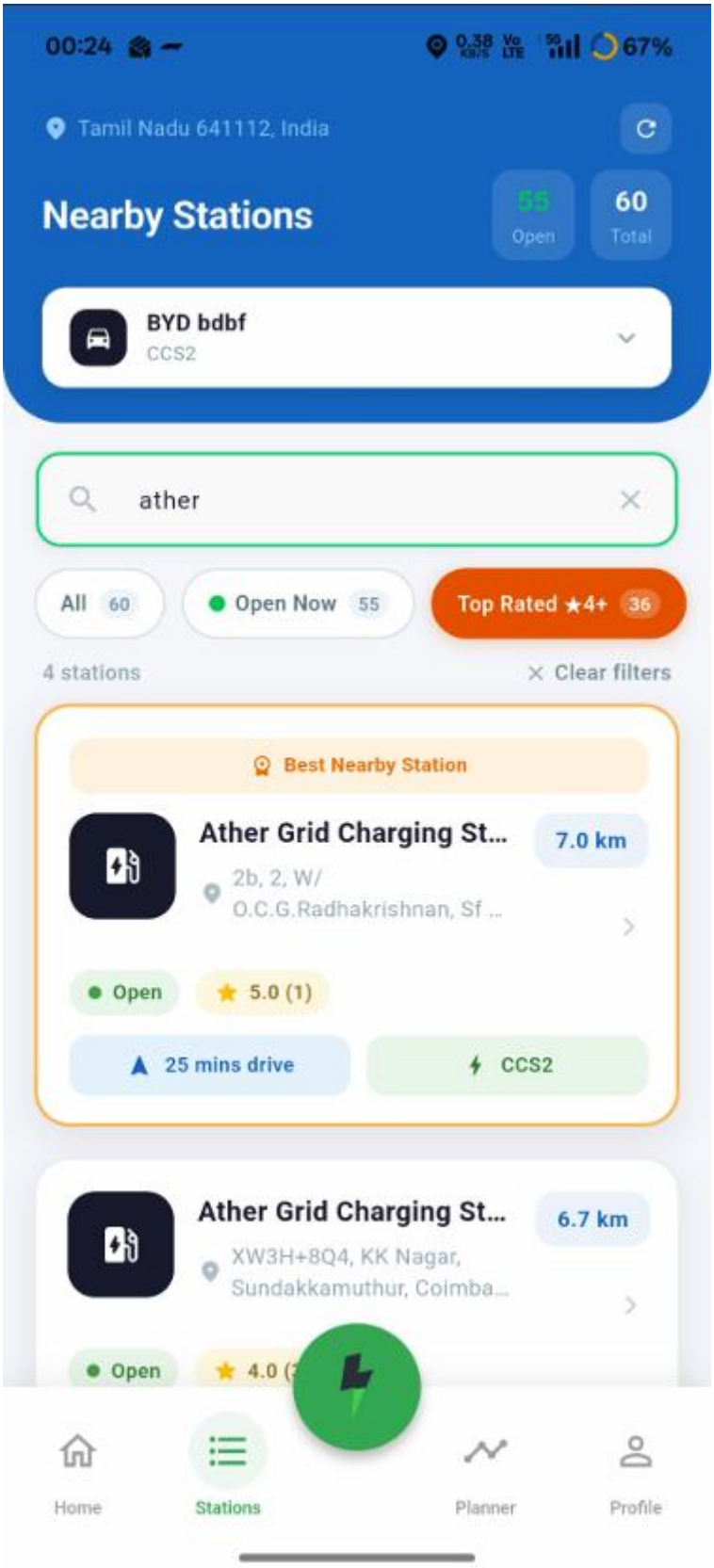

Figure 23. Search filter applied to find specific charging network stations.

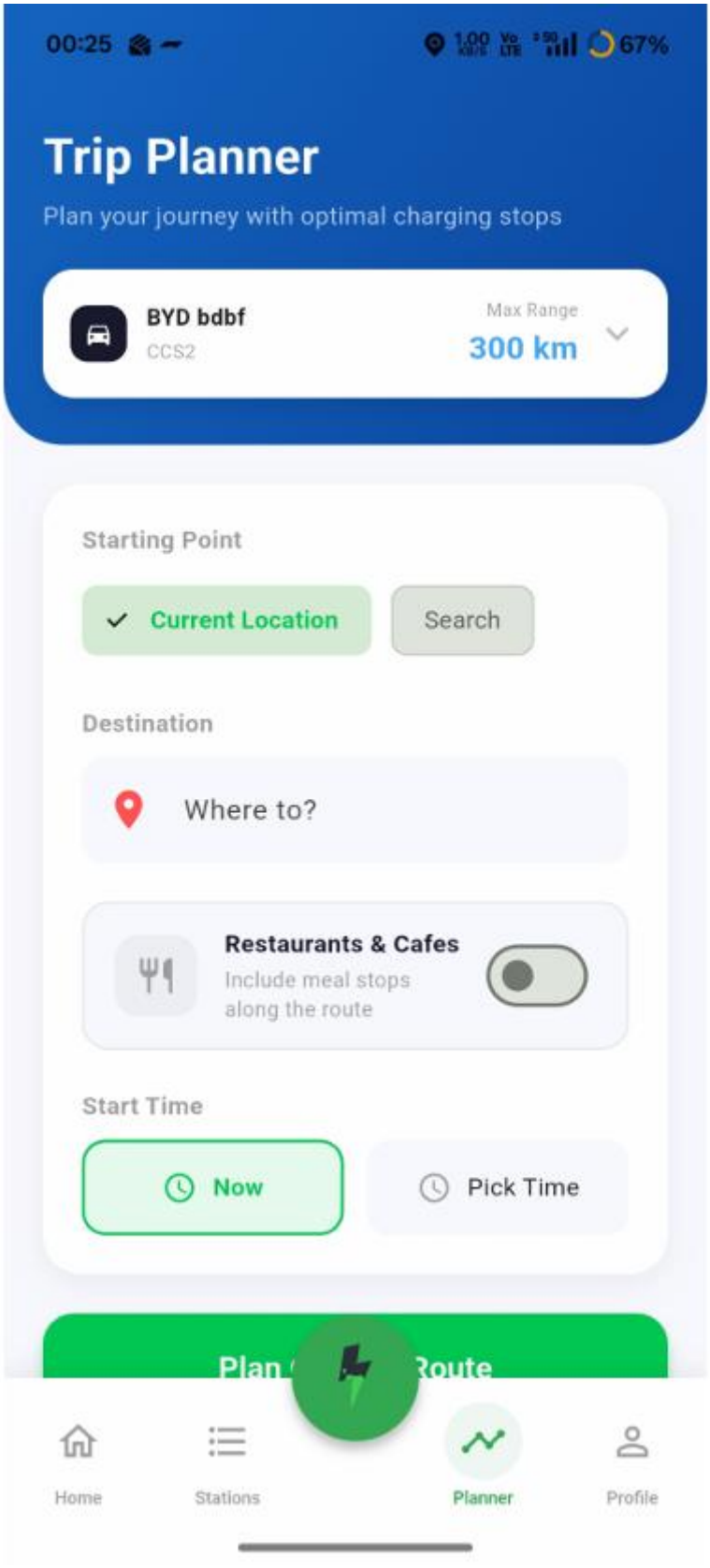

Figure 24. Trip planner screen with route planning options.

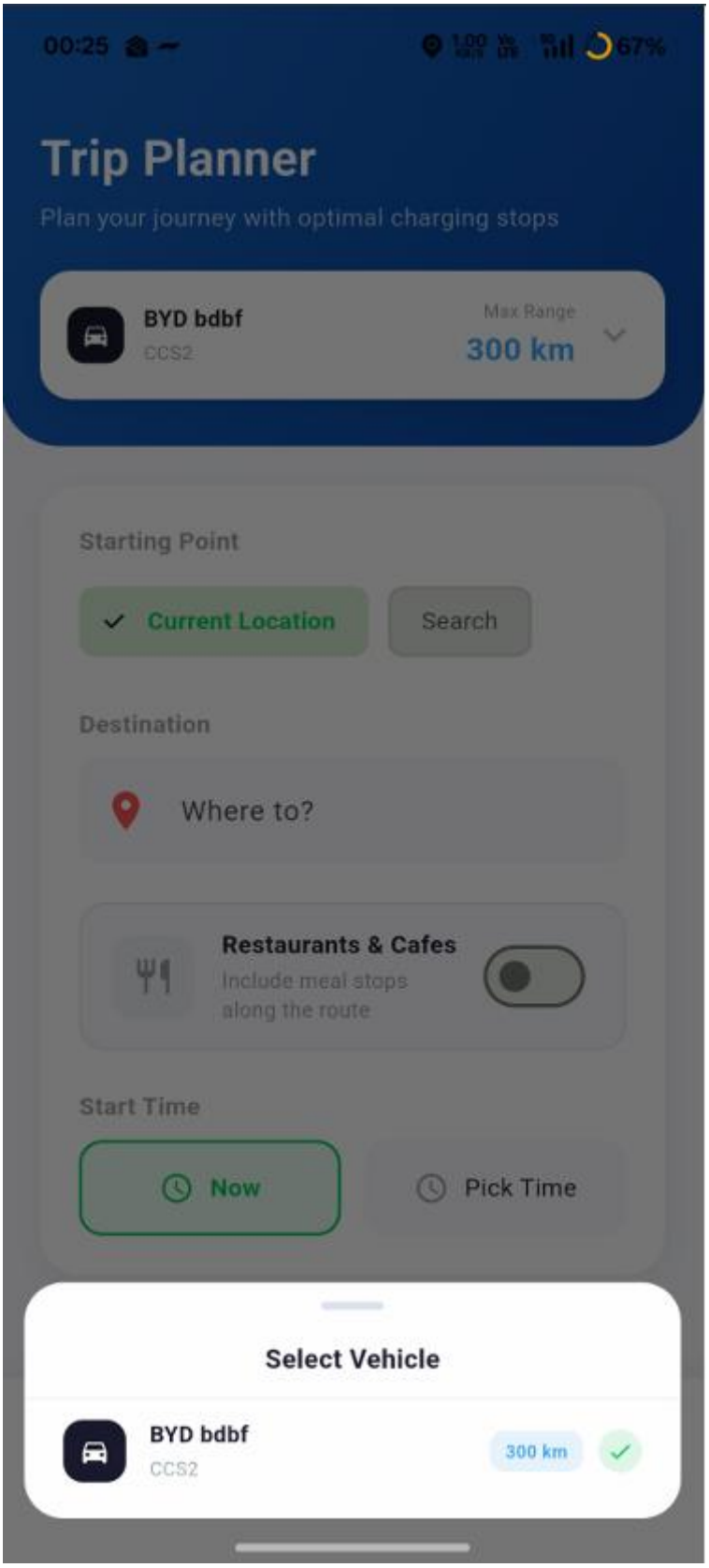

Figure 25. Vehicle selection confirmation for route planning.

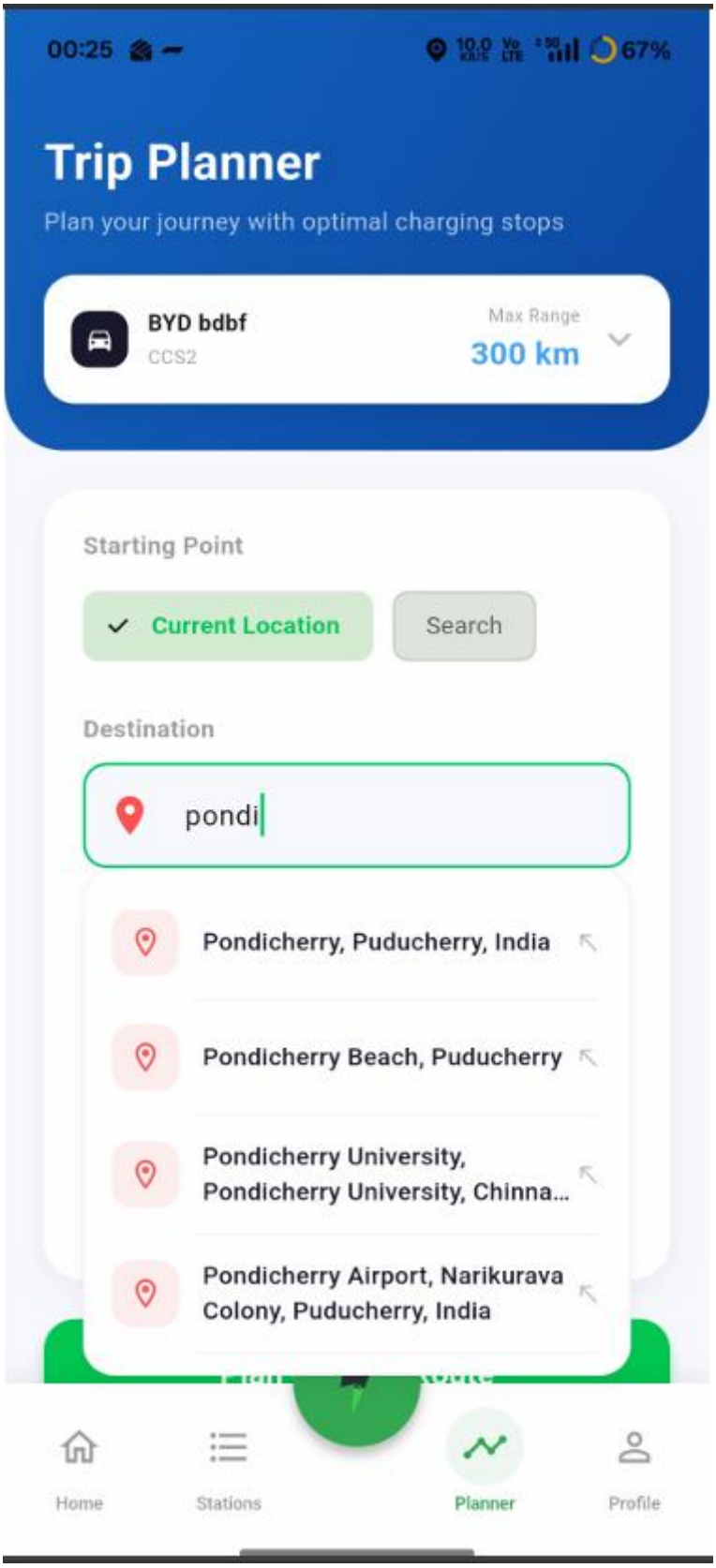

Figure 26. Destination search suggestions in the trip planner module.

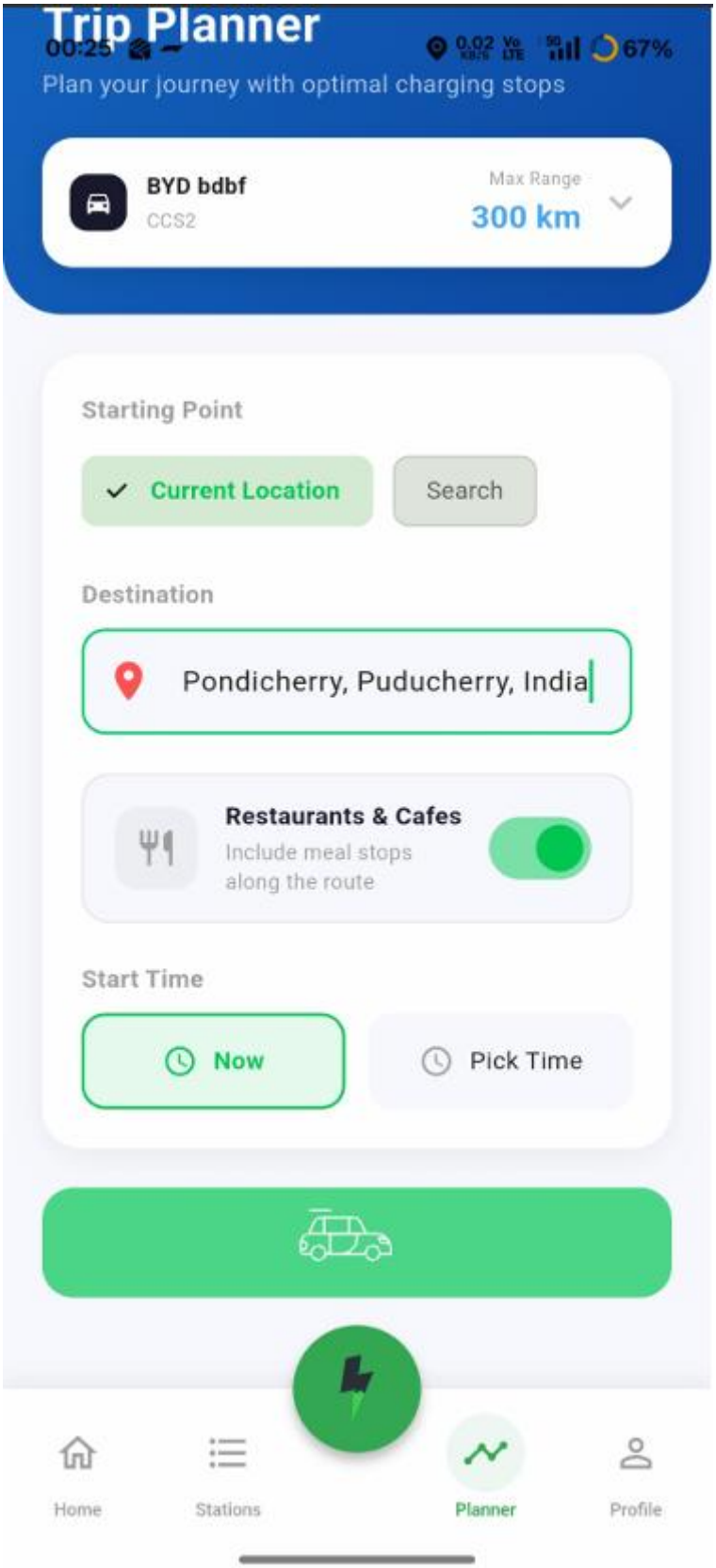

Figure 27. Trip planner interface for planning EV journeys with charging stops.

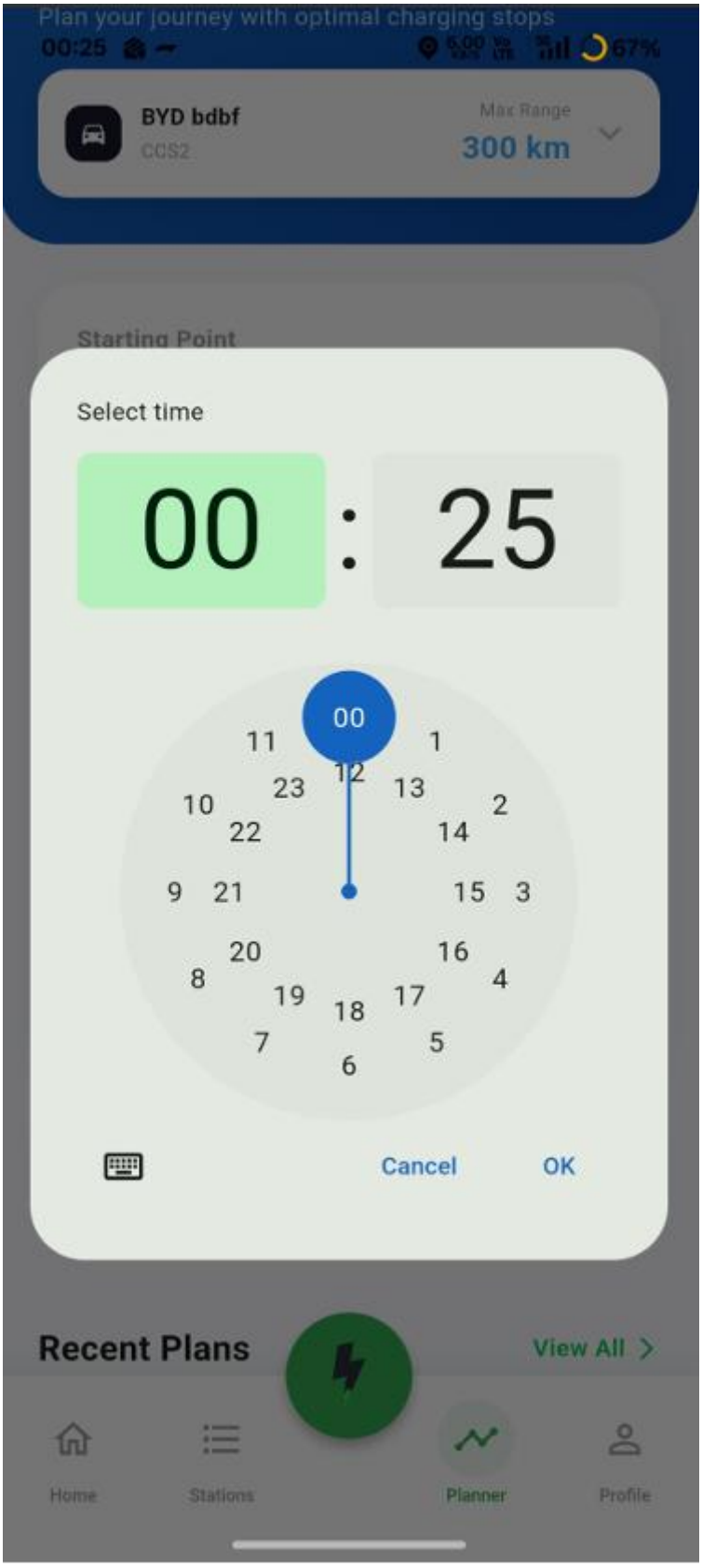

Figure 28. Time selection interface for scheduling trip departure time.

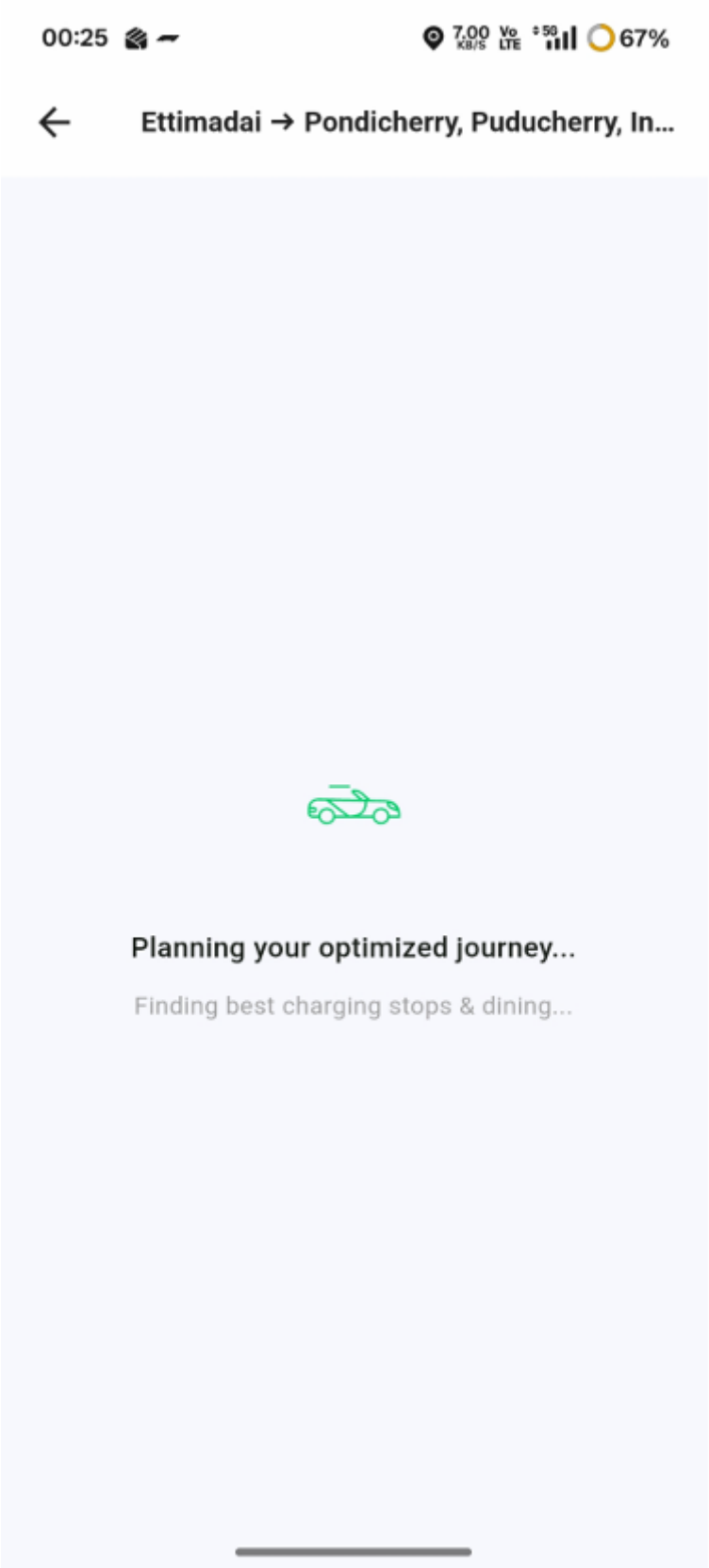

Figure 29. Route optimization loading screen for finding best charging stops.

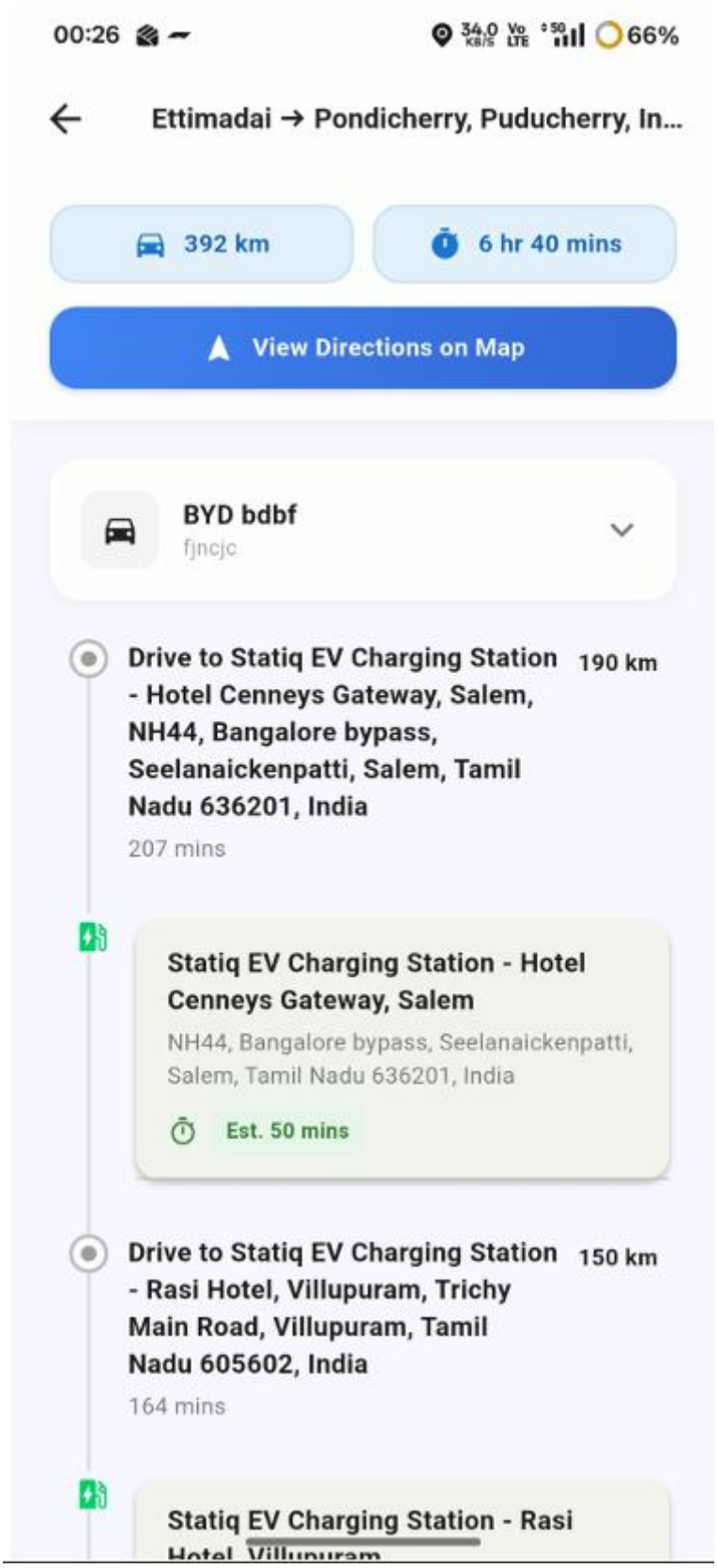

Figure 30. Planned trip summary showing charging stops along the route.

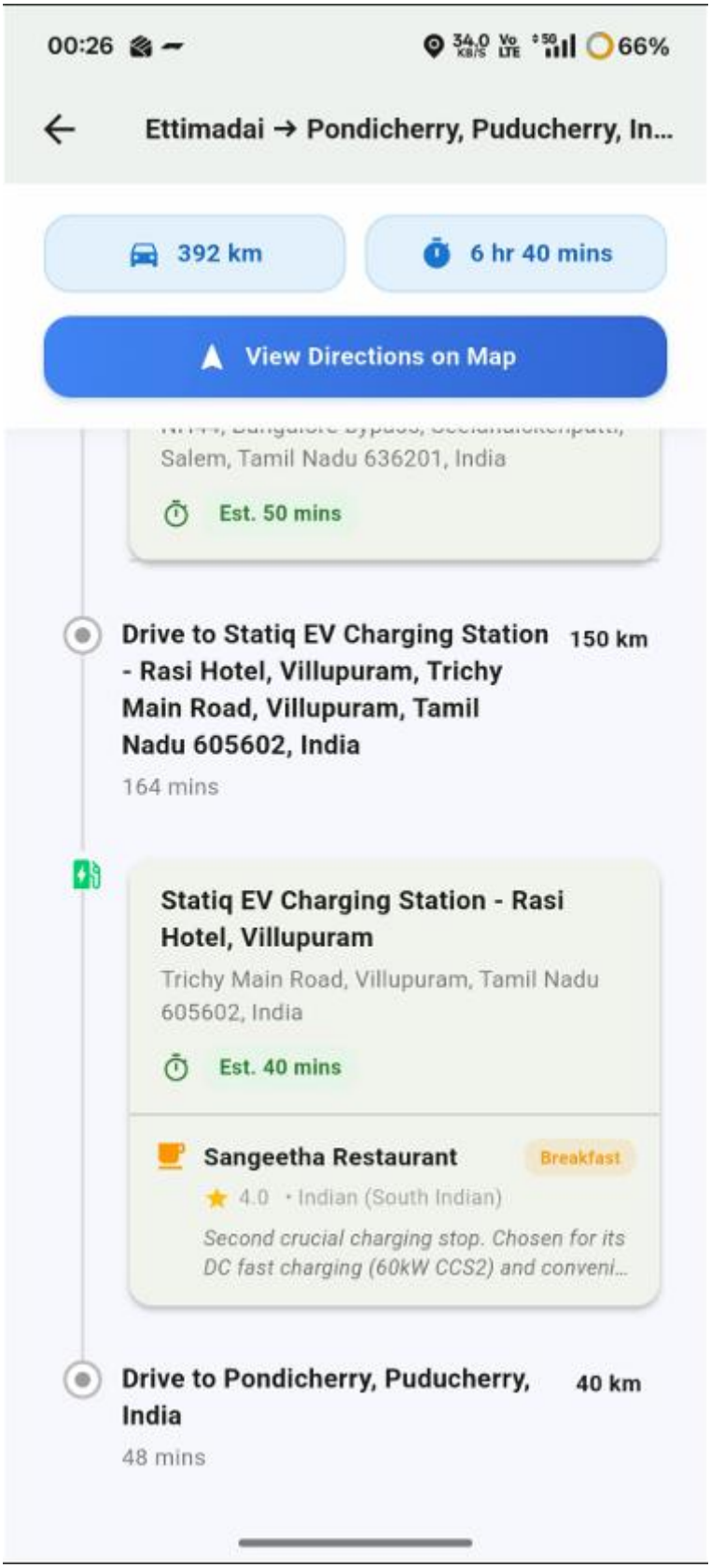

Figure 31. Planned trip summary showing charging stops along the route - cont.

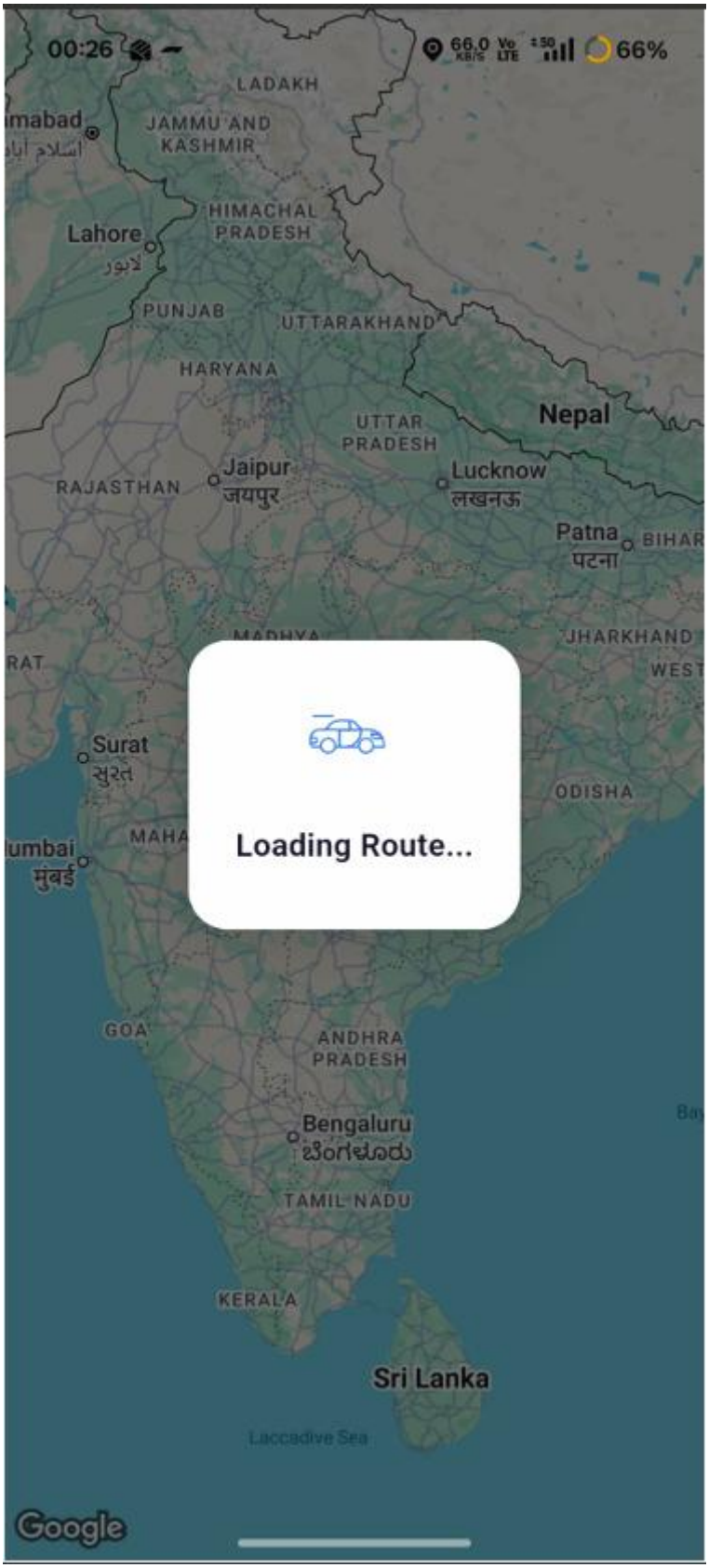

Figure 32. Route loading screen while calculating optimized navigation.

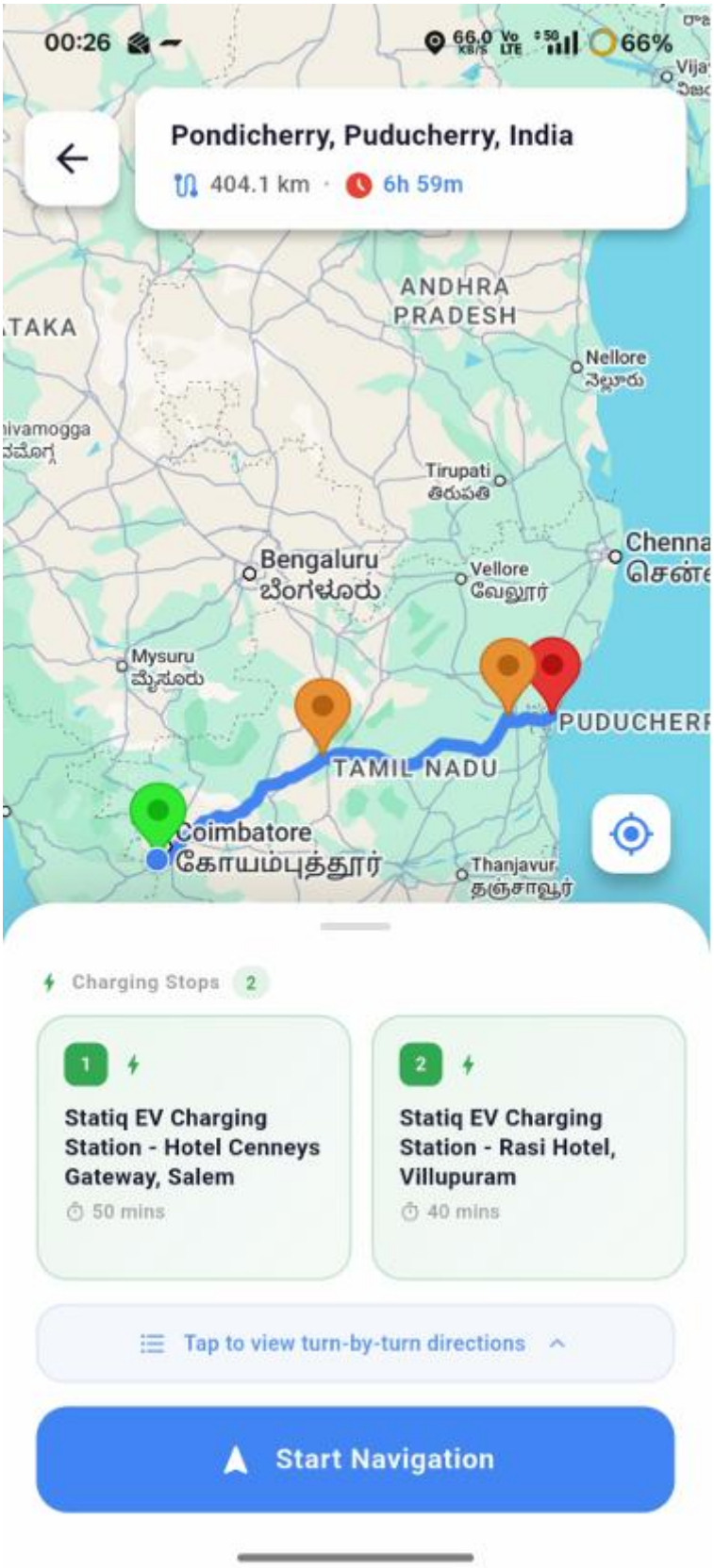

Figure 33. Route overview showing charging stop locations between origin and destination.

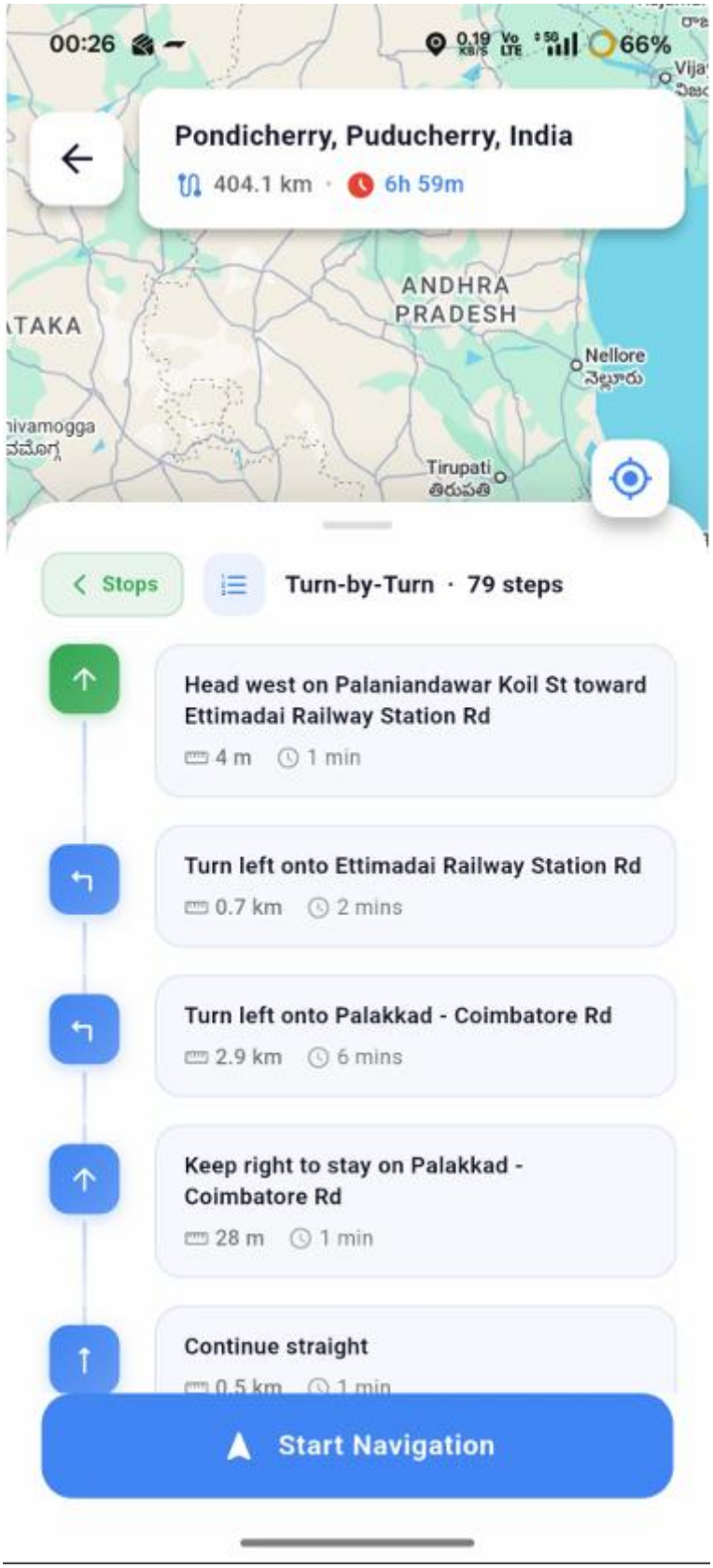

Figure 34. Turn-by-turn route directions for the EV trip.

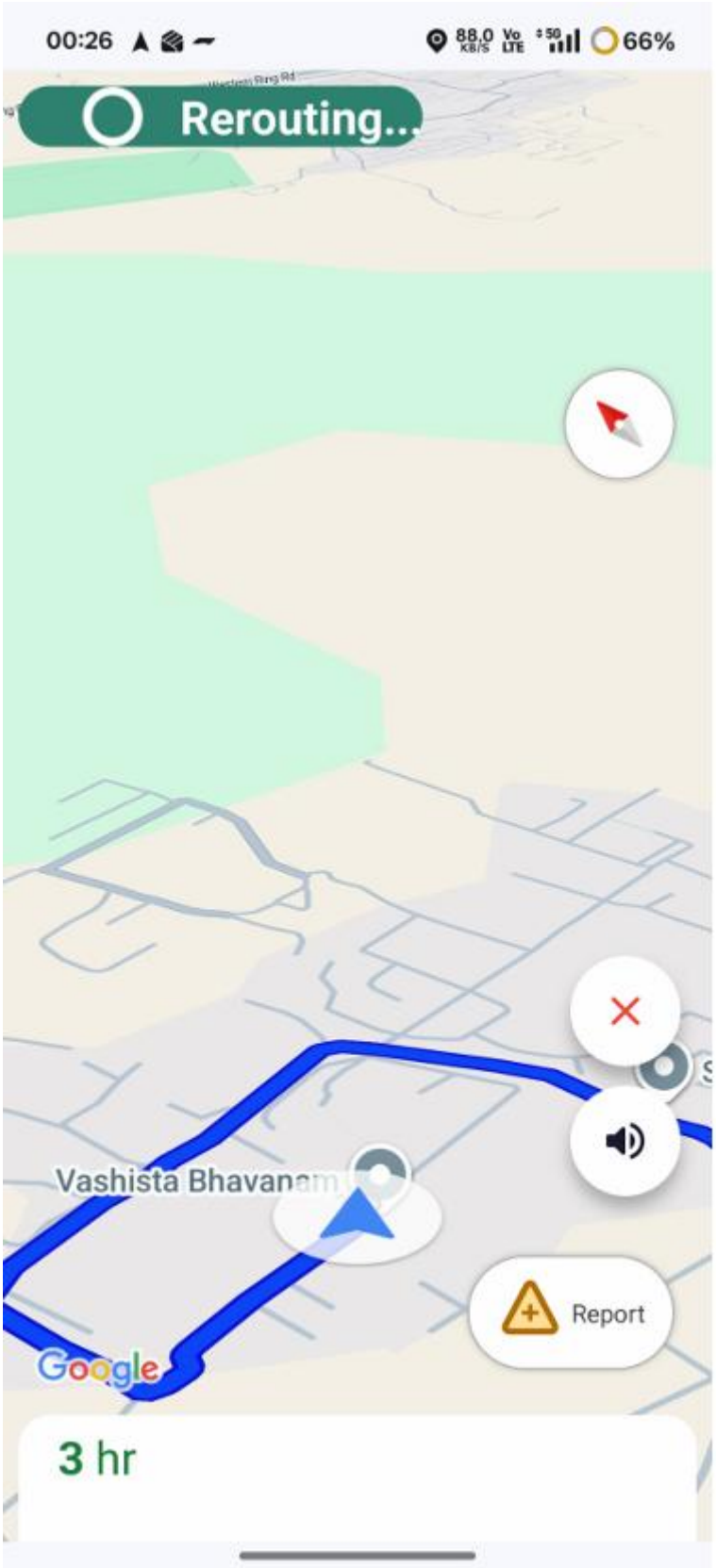

Figure 35. Application settings menu with appearance and account options.

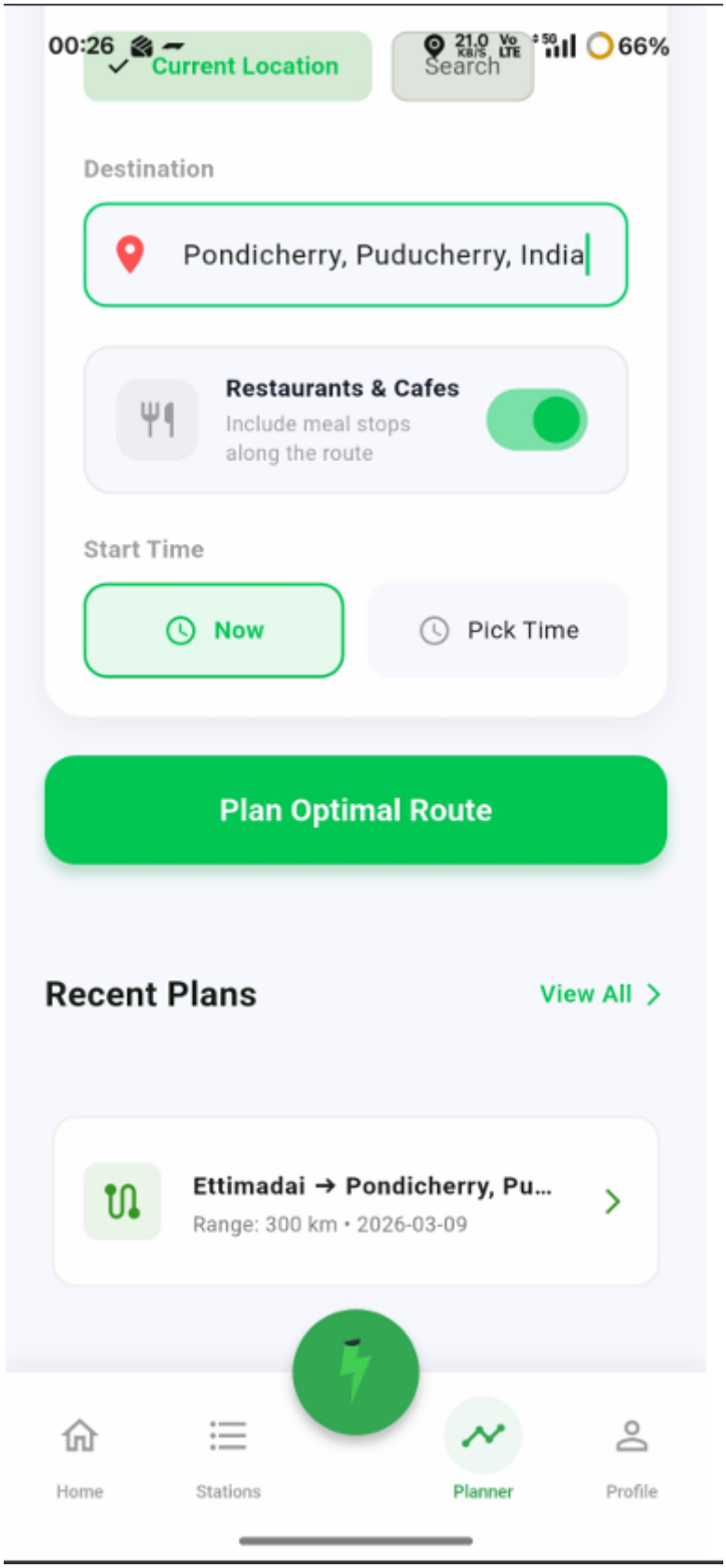

Figure 36. Recent planned trips displayed in the trip planner module.

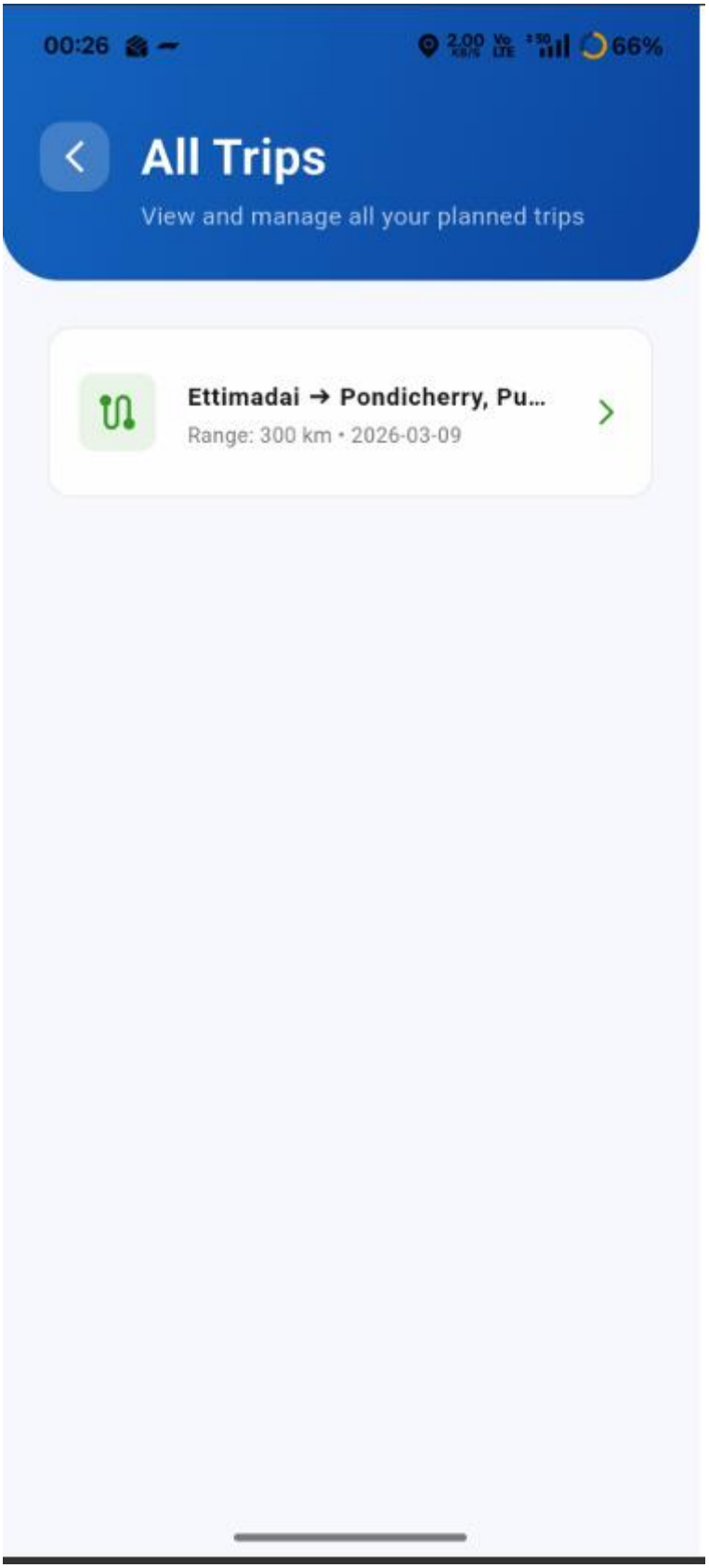

Figure 37. Trip history page displaying saved and completed trips.

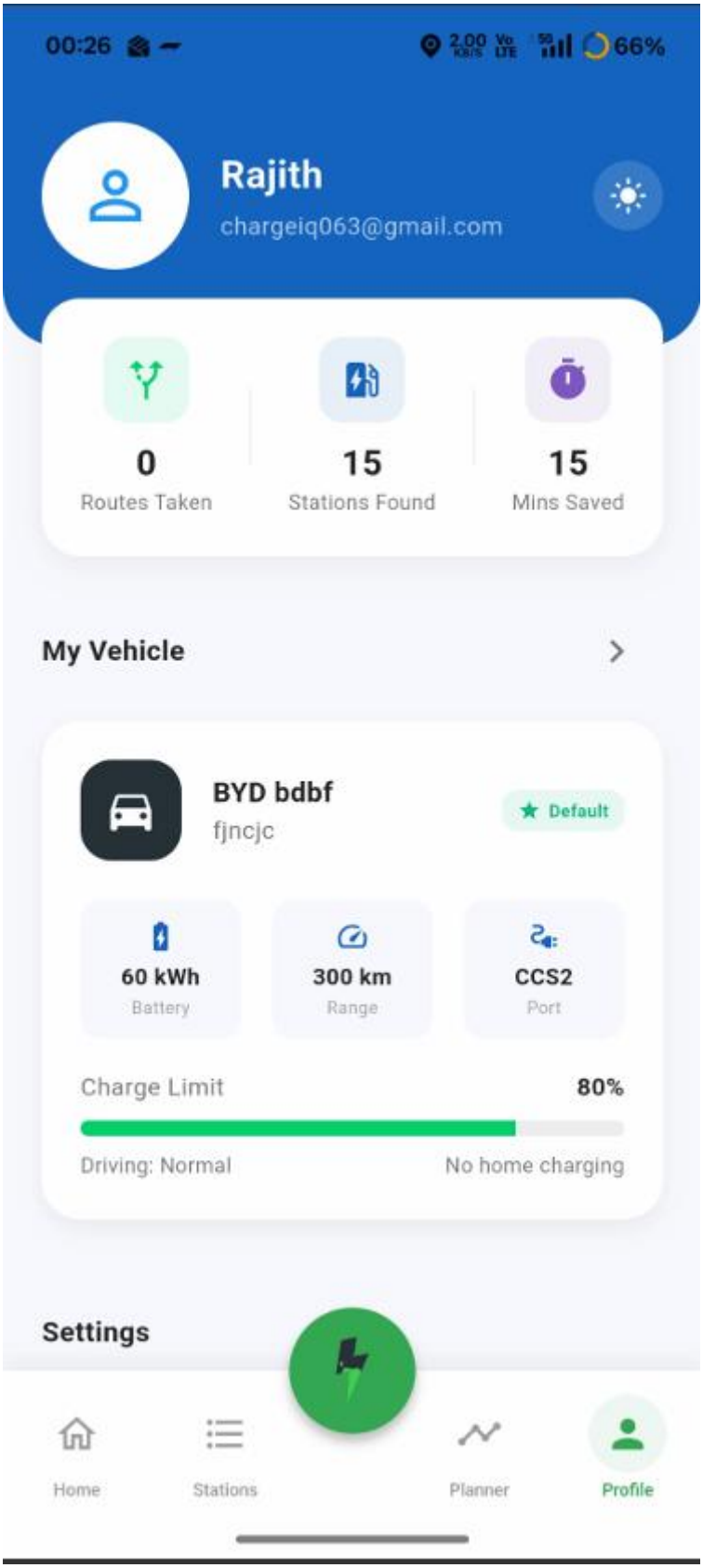

Figure 38. User profile dashboard displaying vehicle information and usage statistics.

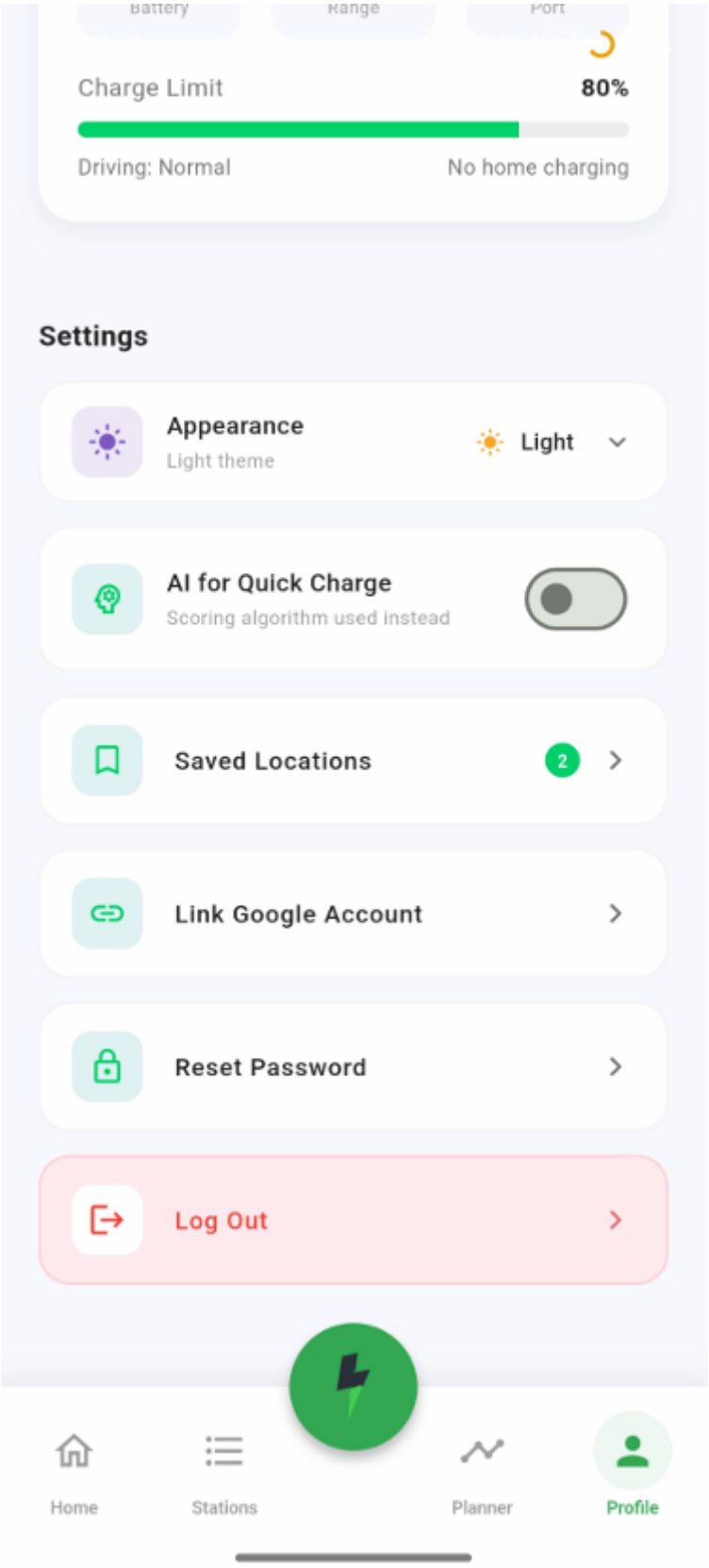

Figure 39. Application settings menu with appearance and account options.

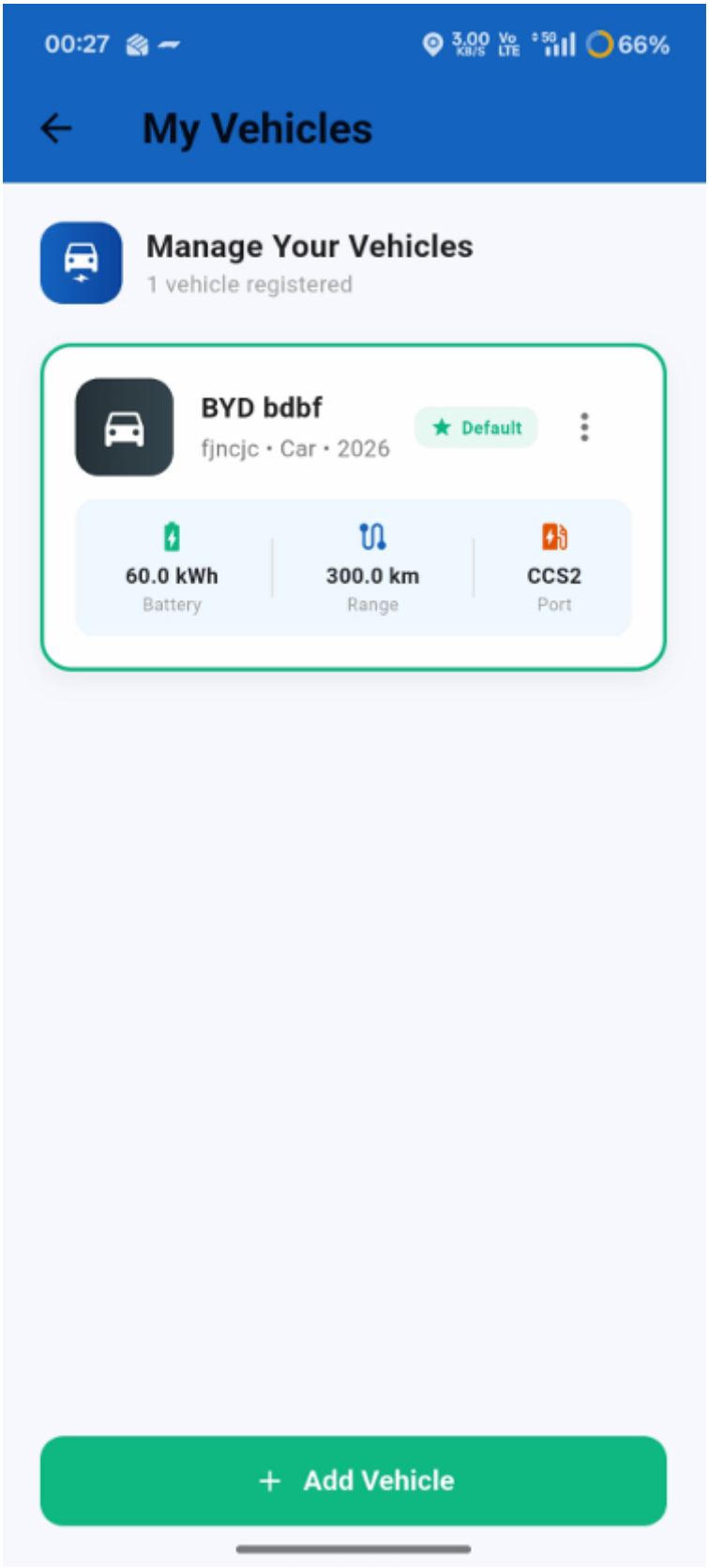

Figure 40. Vehicle management screen showing registered EV details.

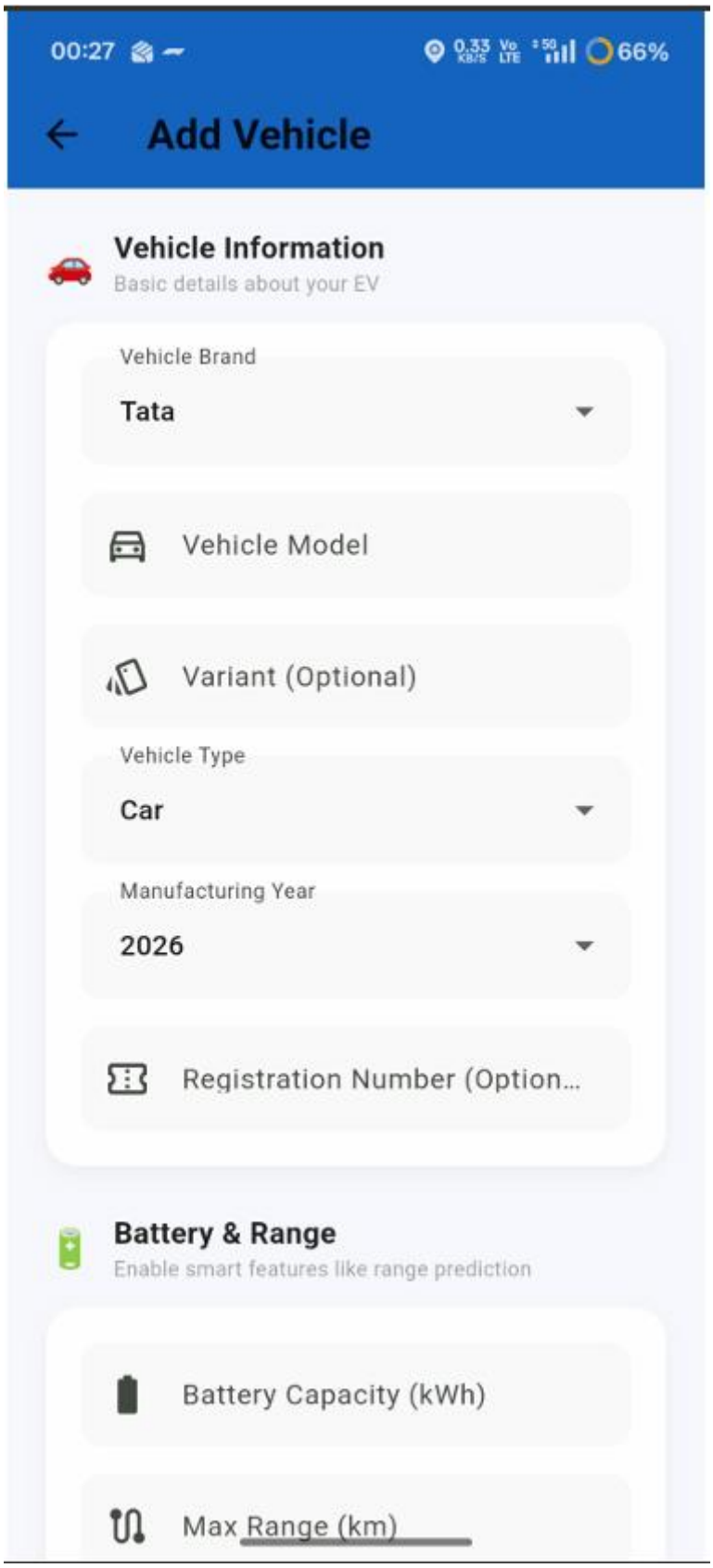

Figure 41. Add vehicle screen with smart charging personalization settings.

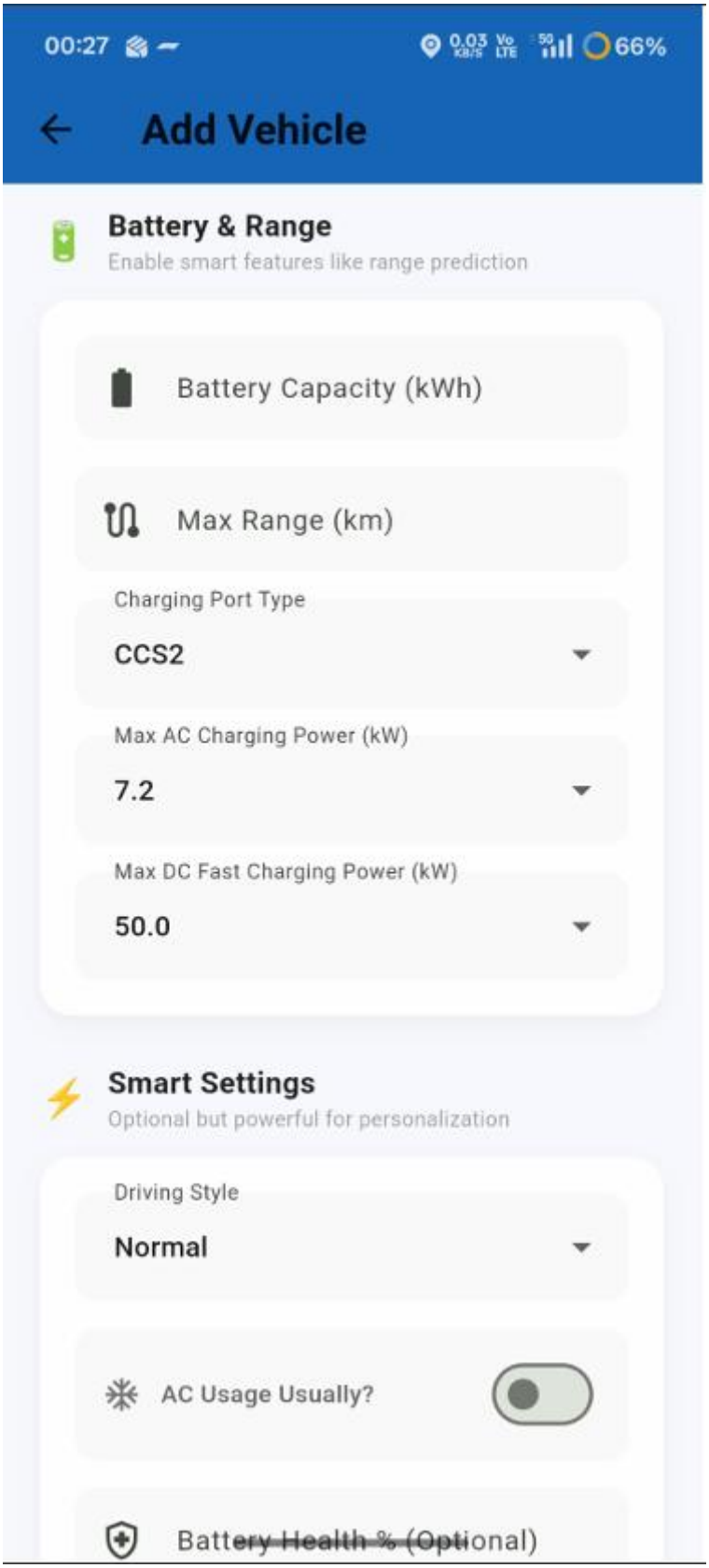

Figure 42. Add vehicle screen with smart charging personalization settings- cont.

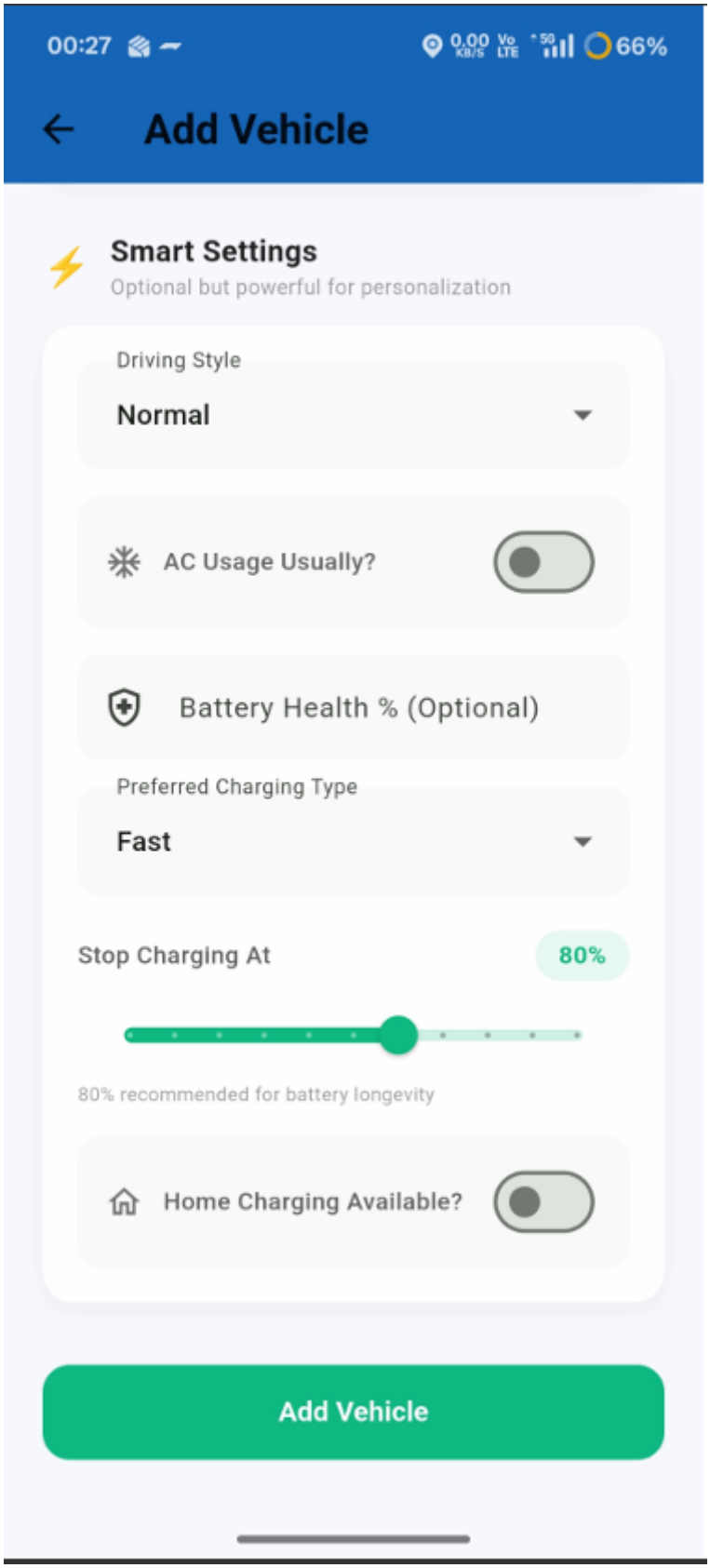

Figure 43. Add vehicle screen with smart charging personalization settings - cont.

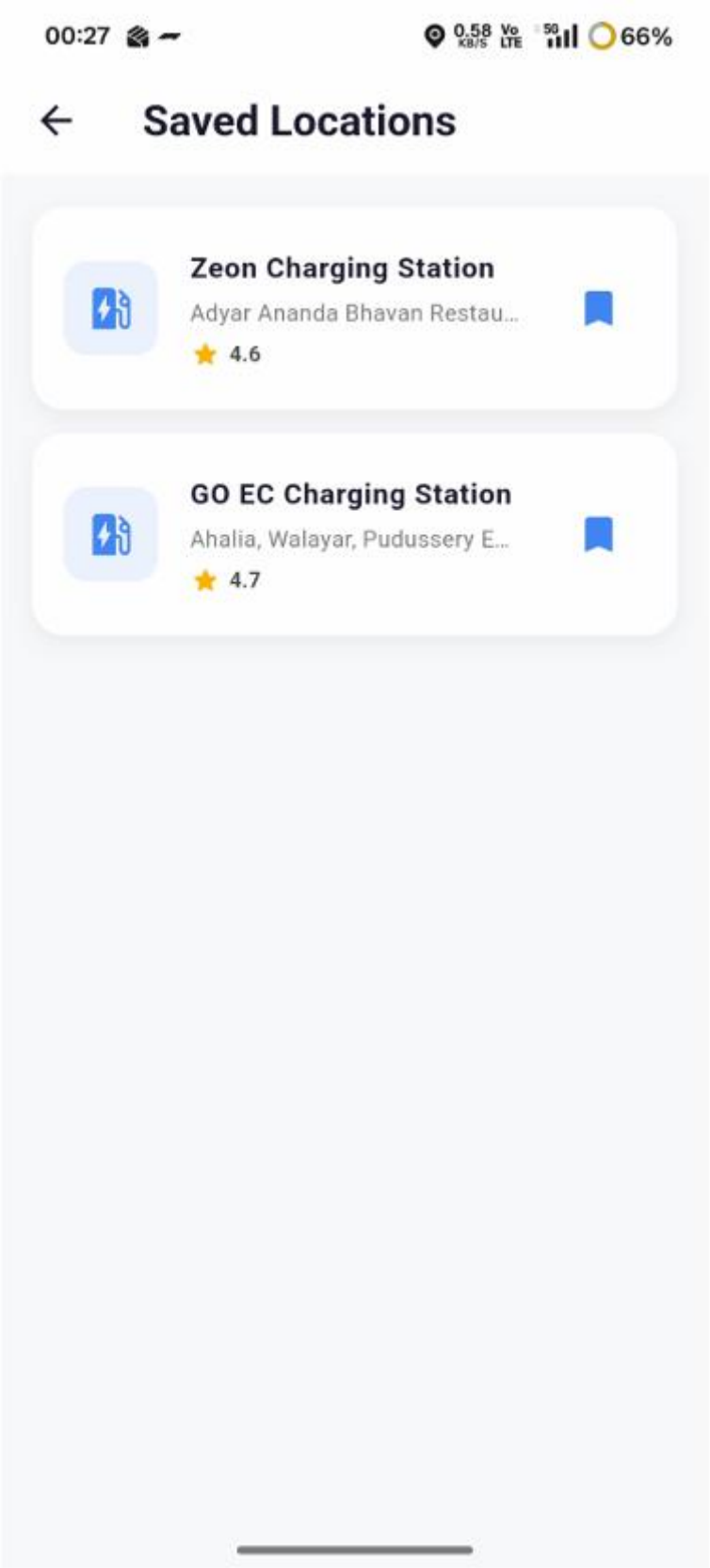

Figure 44. Saved charging station locations for quick access.

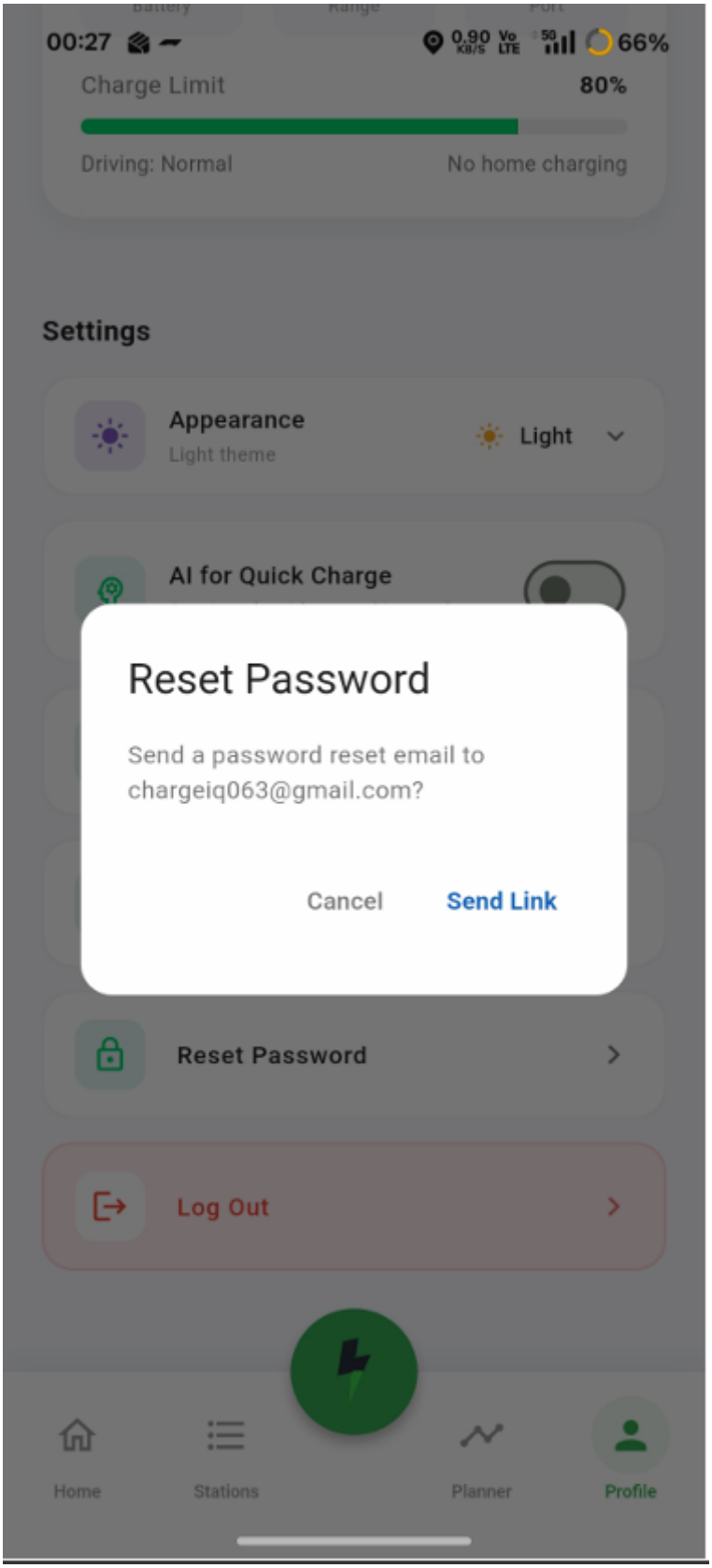

Figure 45. Password reset confirmation dialog in the settings section.

DARK MODE

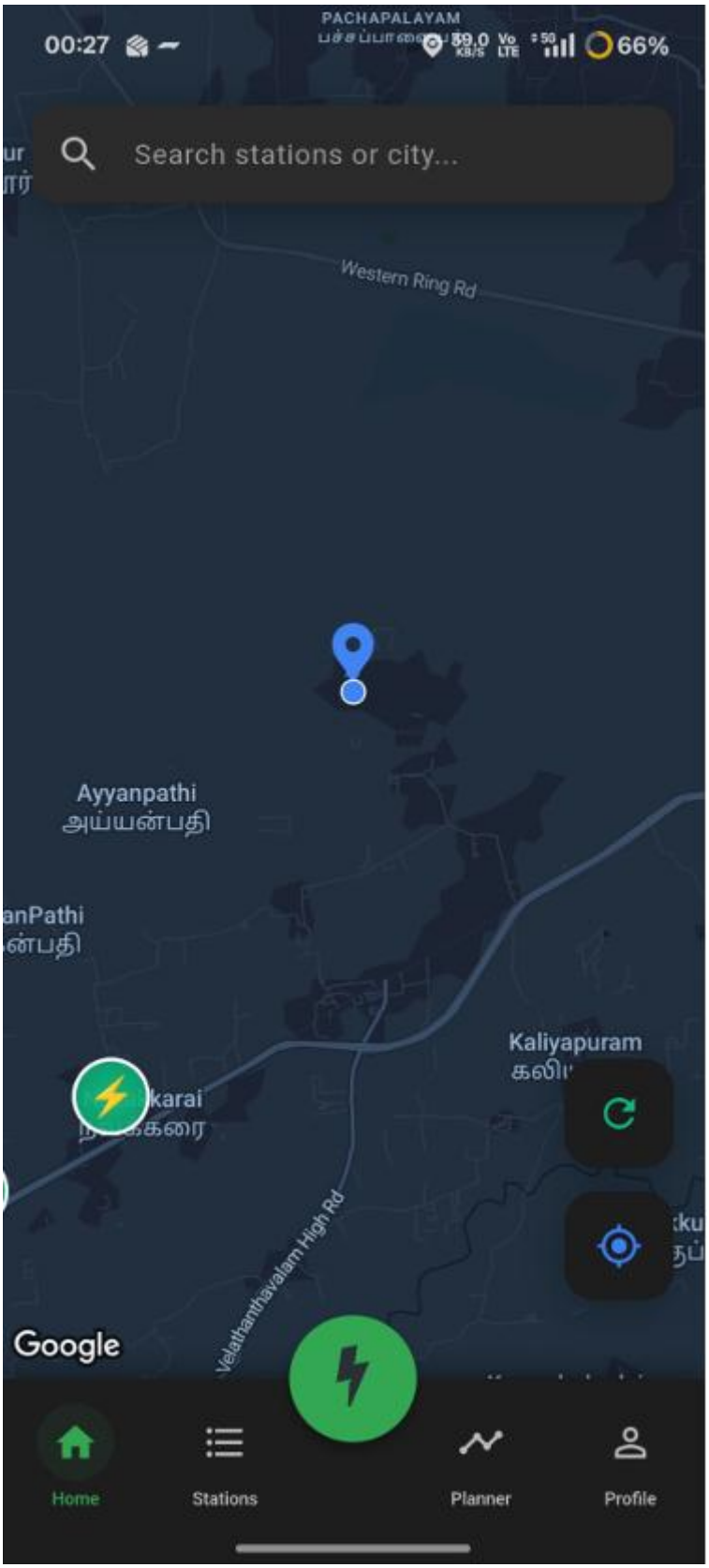

Figure 46. Map view in dark mode

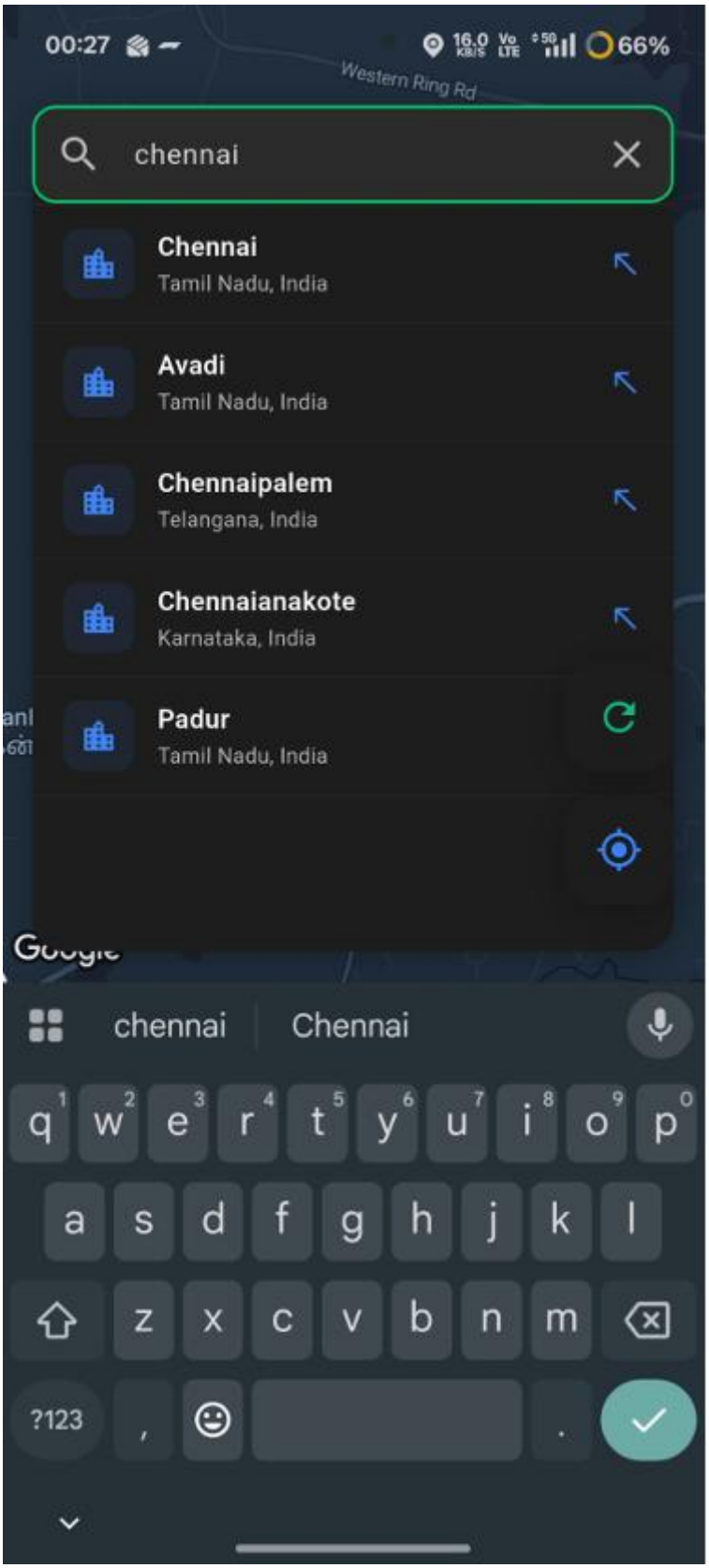

Figure 47. Search bar suggestions in dark mode

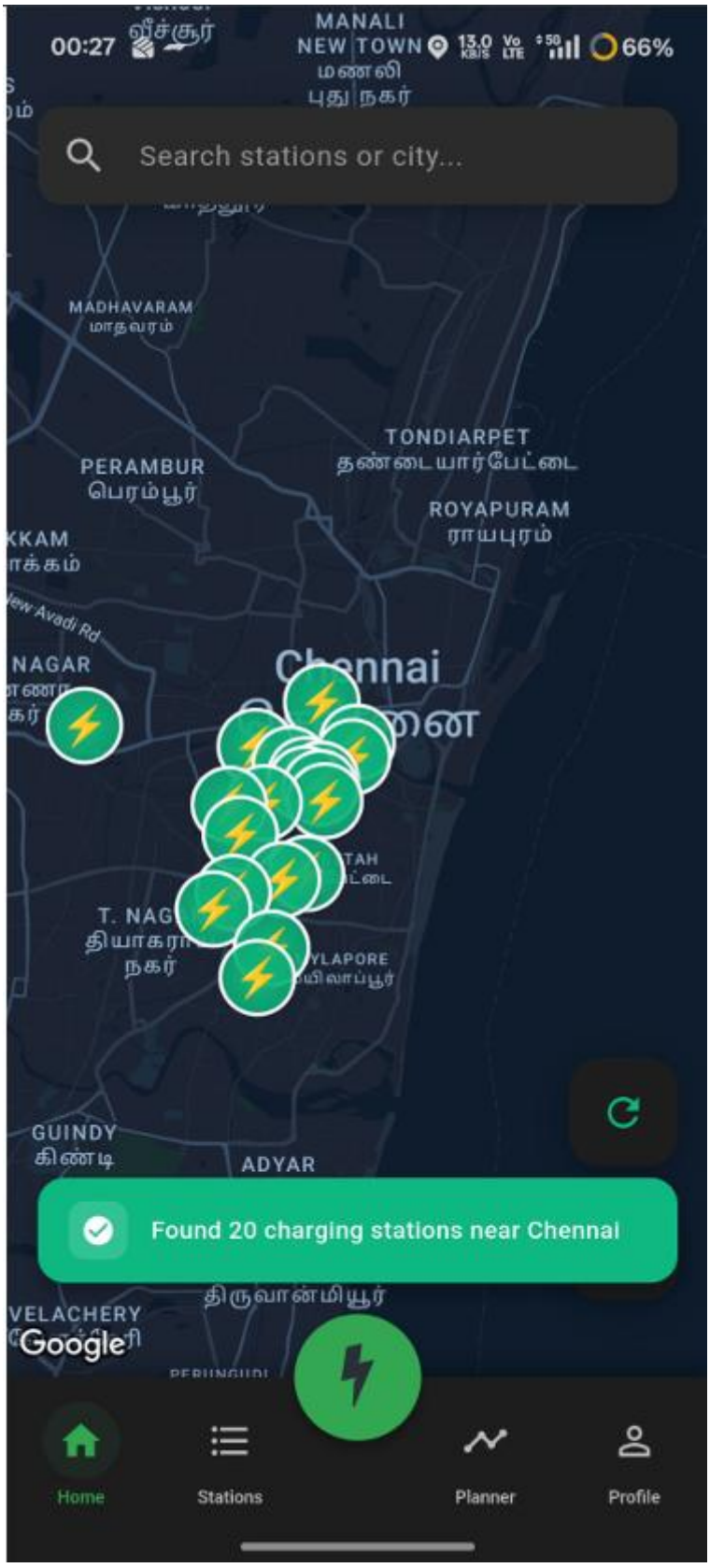

Figure 48. Map displaying multiple EV charging stations near Chennai in dark mode.

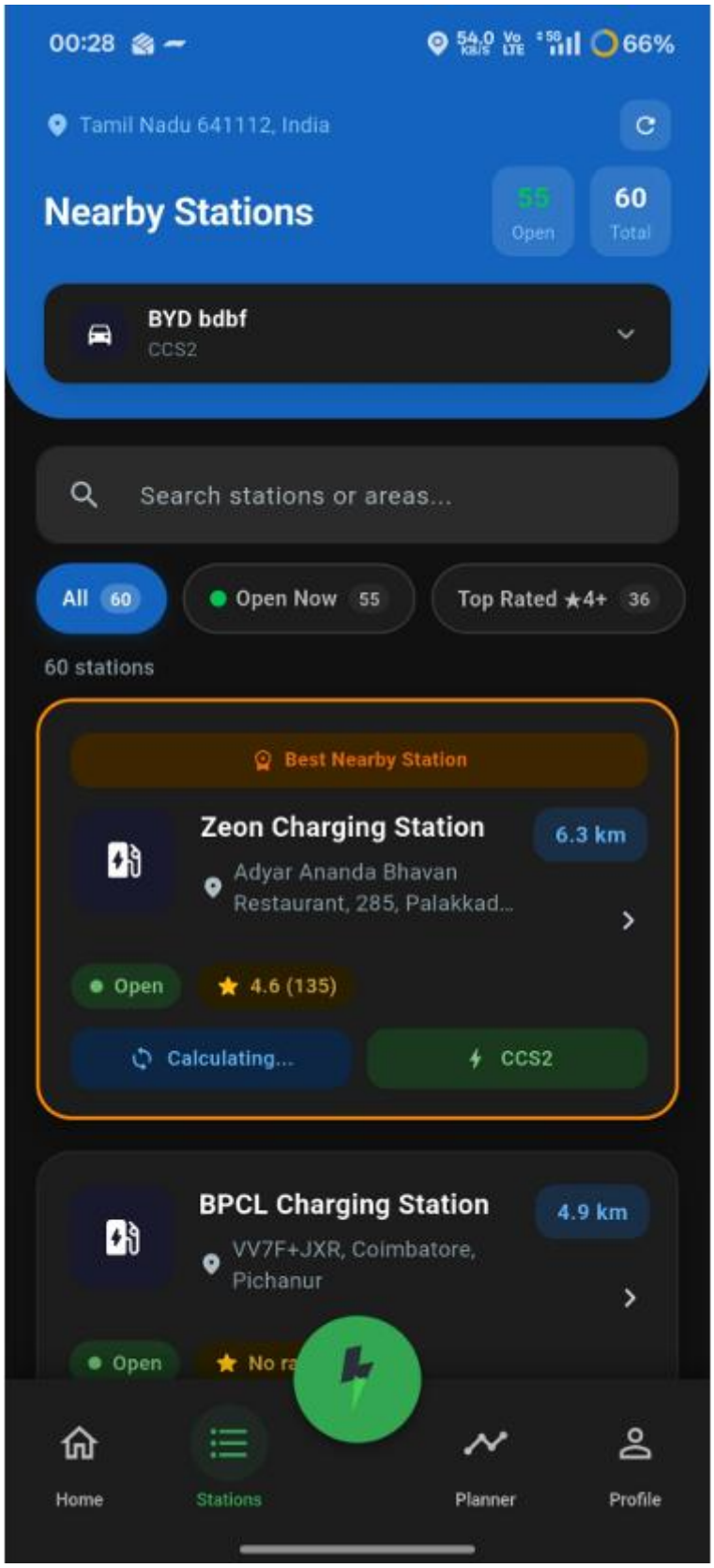

Figure 49. Nearby charging station list in dark mode interface.

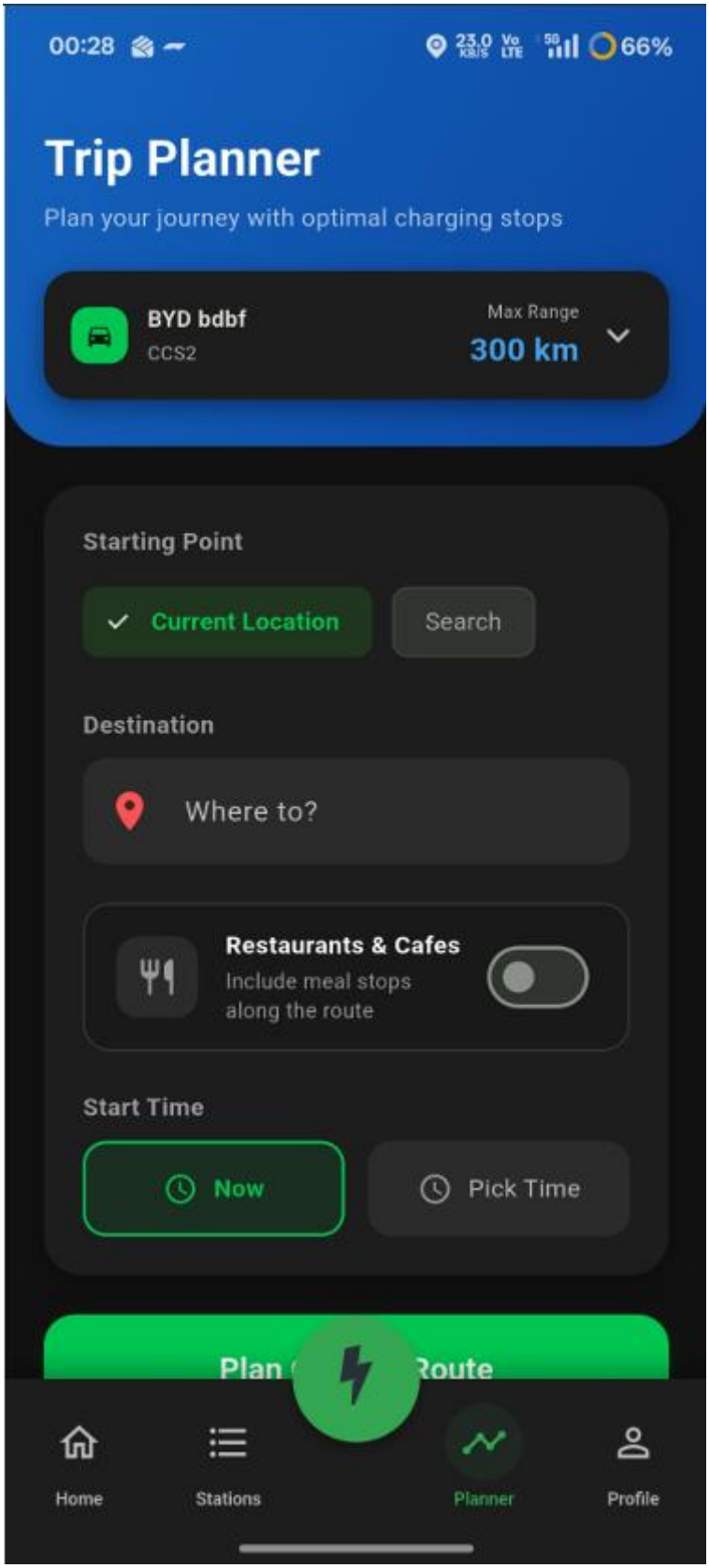

Figure 50. Trip planner interface displayed in dark theme.

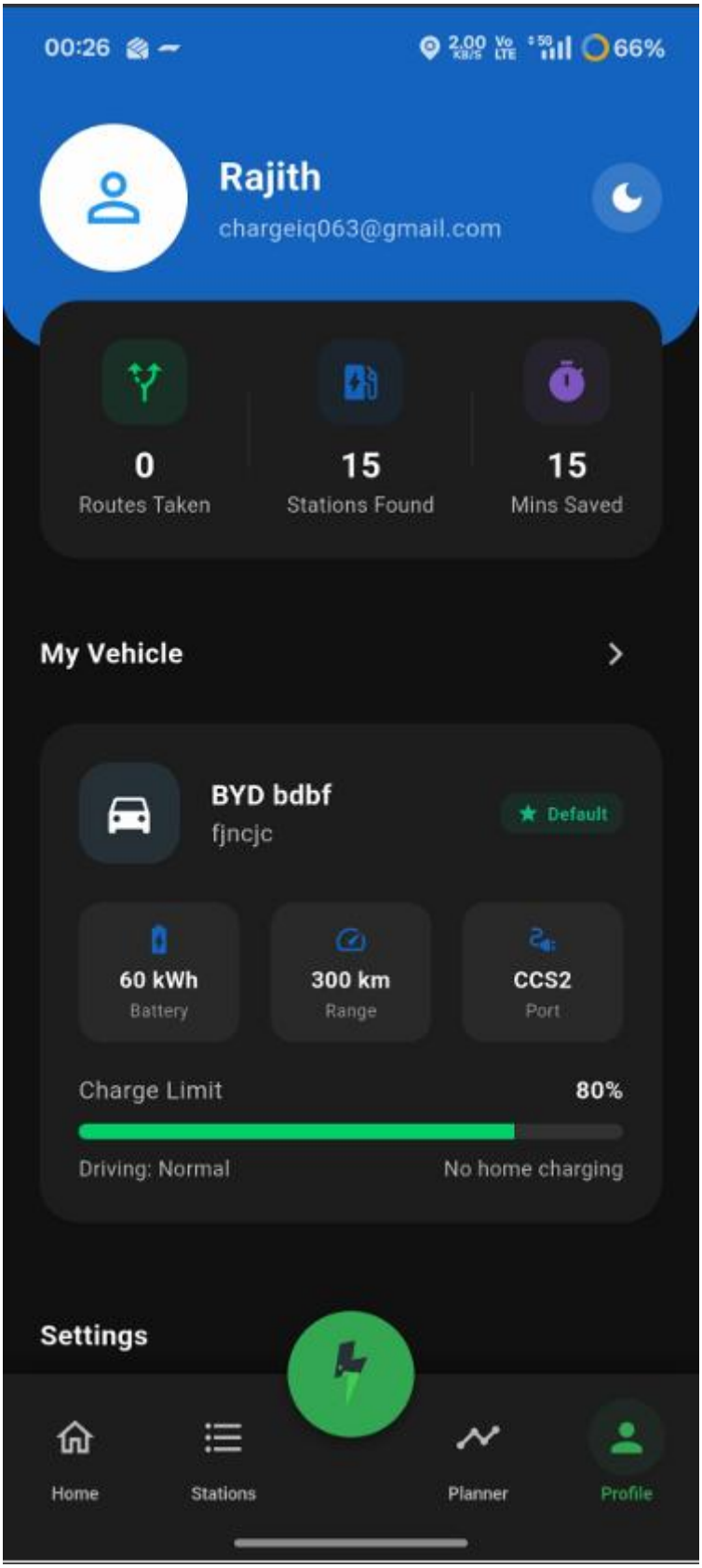

Figure 51. User profile page in dark mode

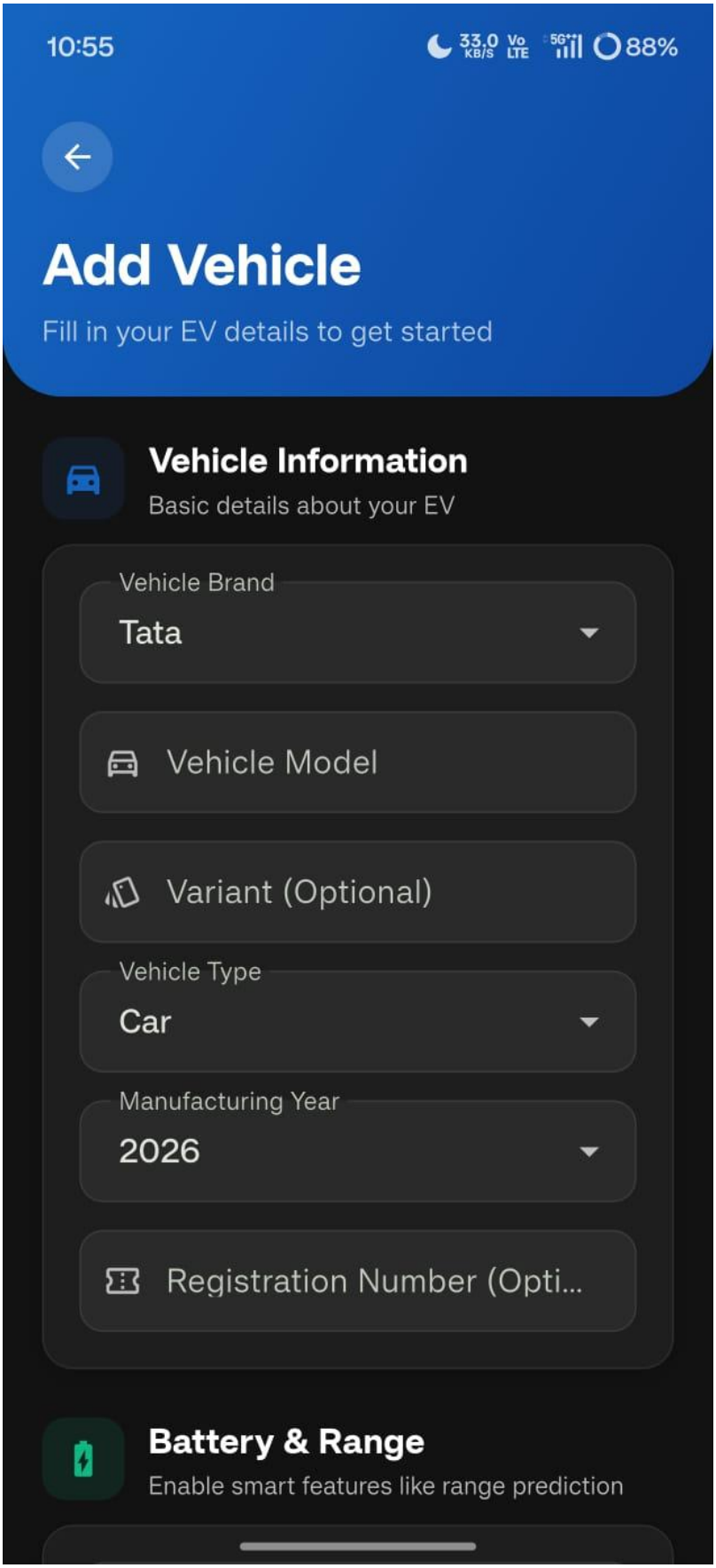

Figure 52. Add vehicle page in dark mode

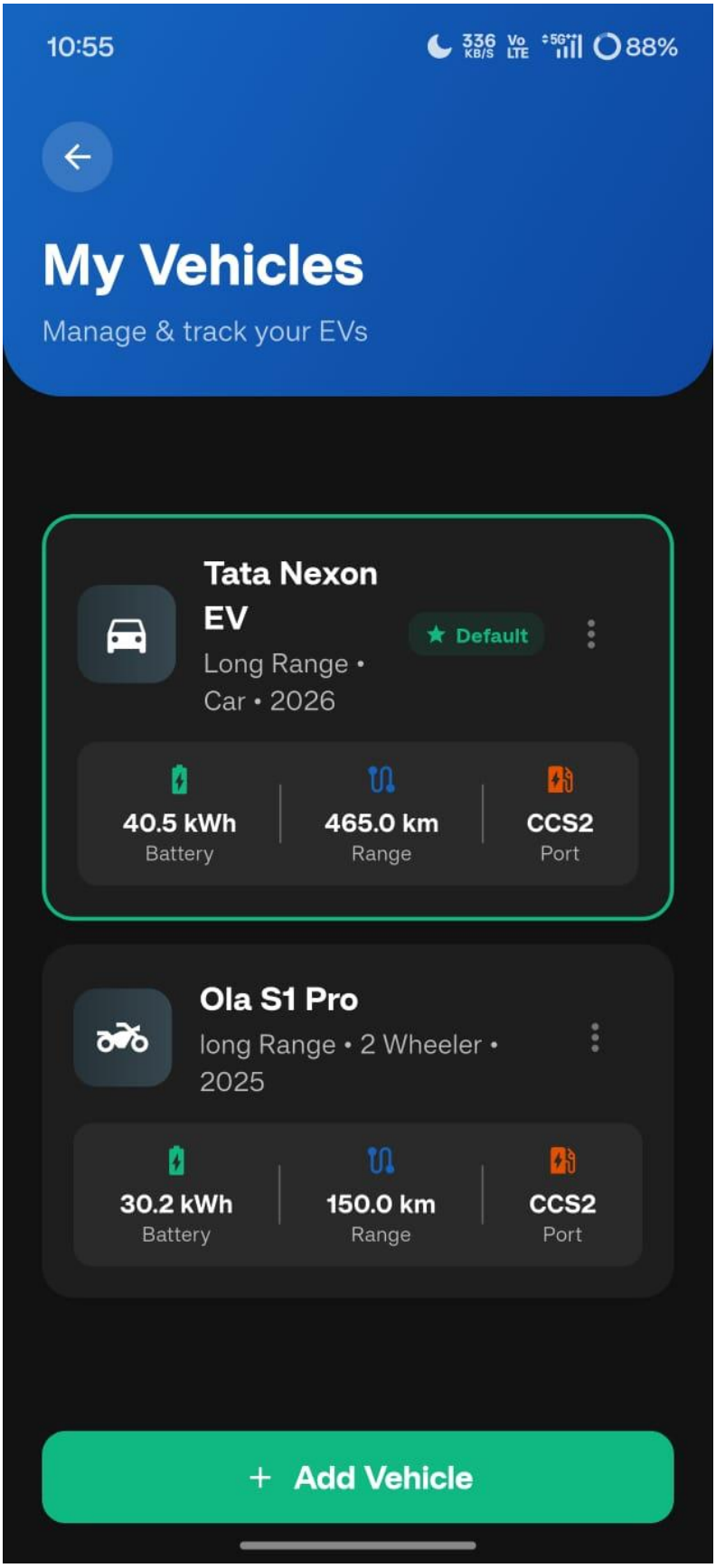

Figure 53. Vehicle management screen in dark mode